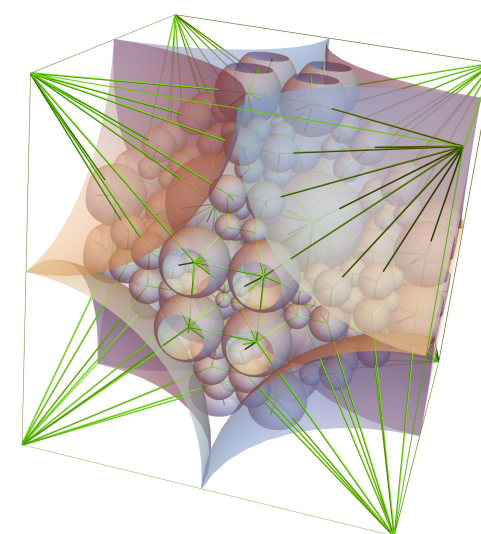
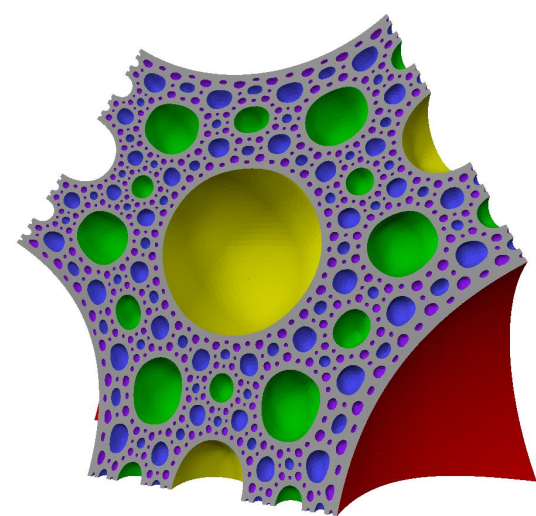


Représentation des empilements polytopaux de sphères à l'aide du modèle BC-IFS



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LIB
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Boris Bordeaux¹, Iván Rasskin², Christian Gentil¹, Jorge Luis Ramírez Alfonsín³, Lionel Garnier¹

¹*LIB – Université de Bourgogne*
²*AZT – Université Technique de Graz*
³*IMAG – Université de Montpellier*

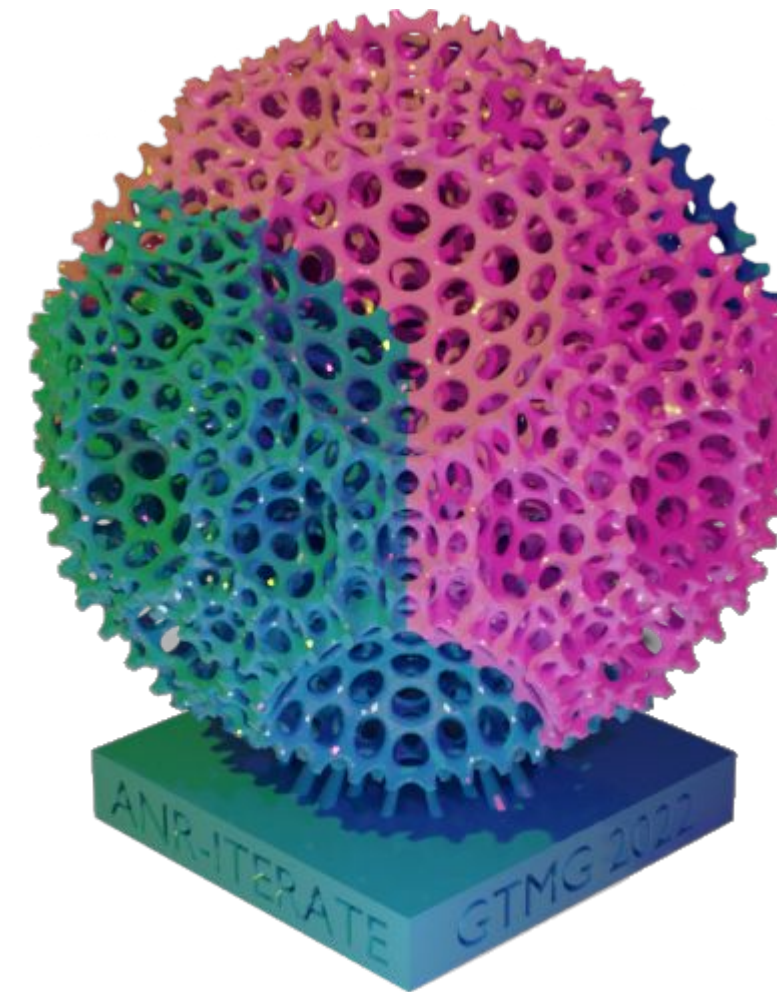
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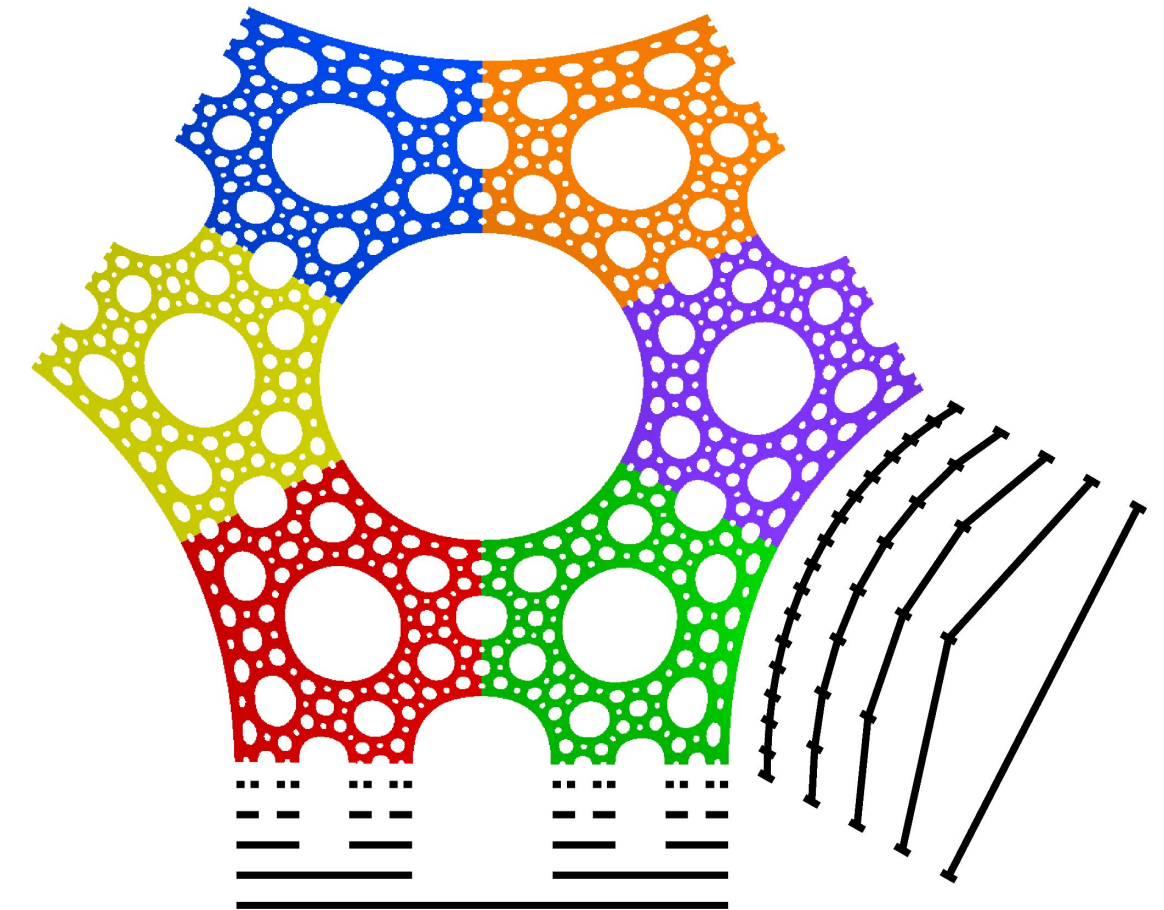
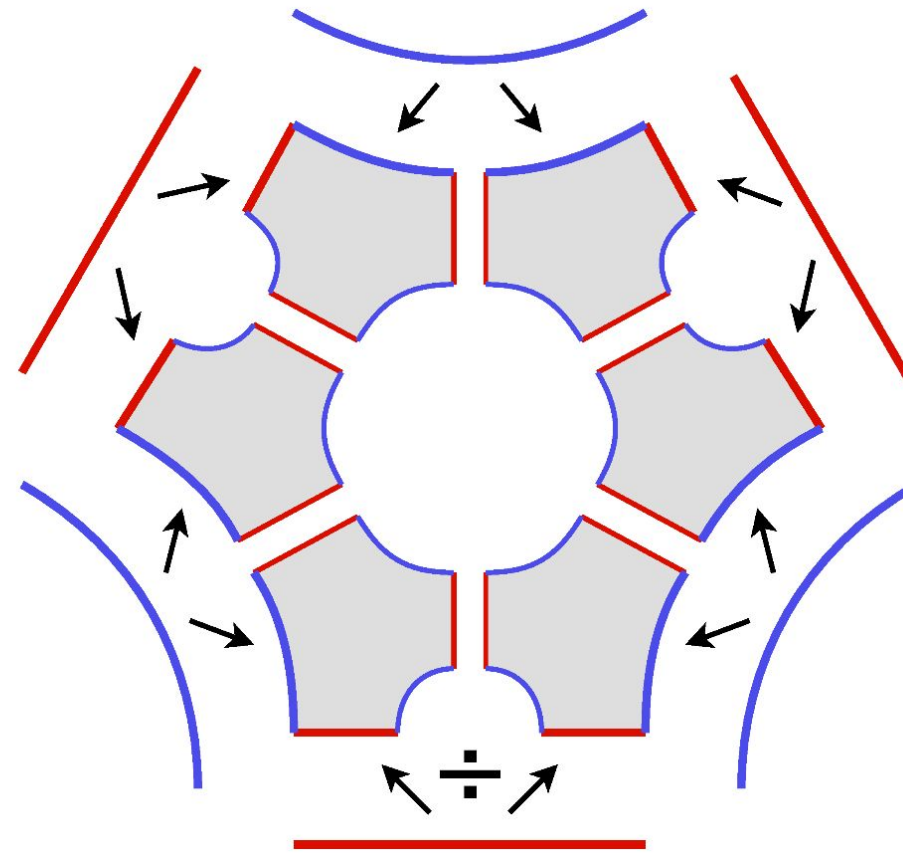
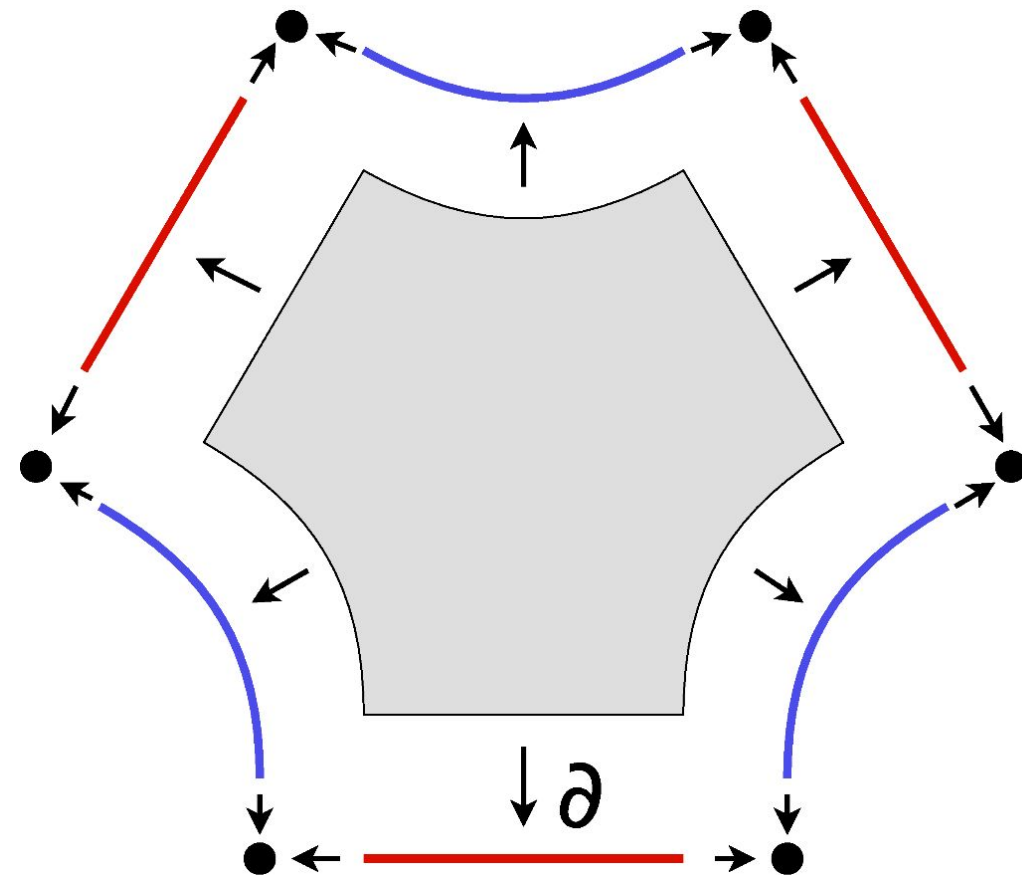
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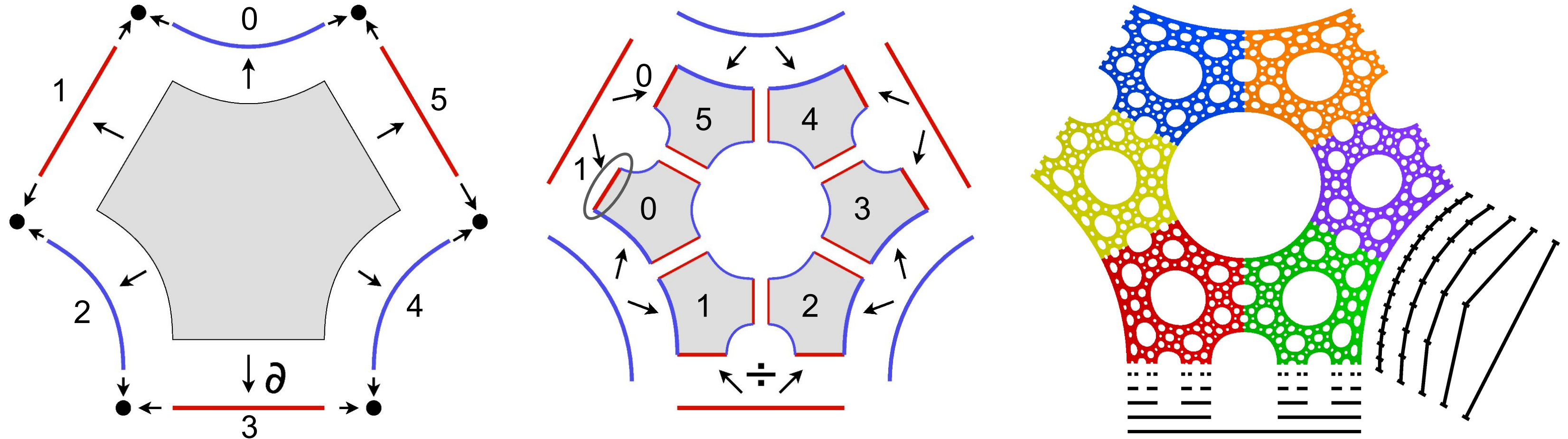
15-16 mars 2023, GTMG - Strasbourg

COLLÈGE DOCTORAL
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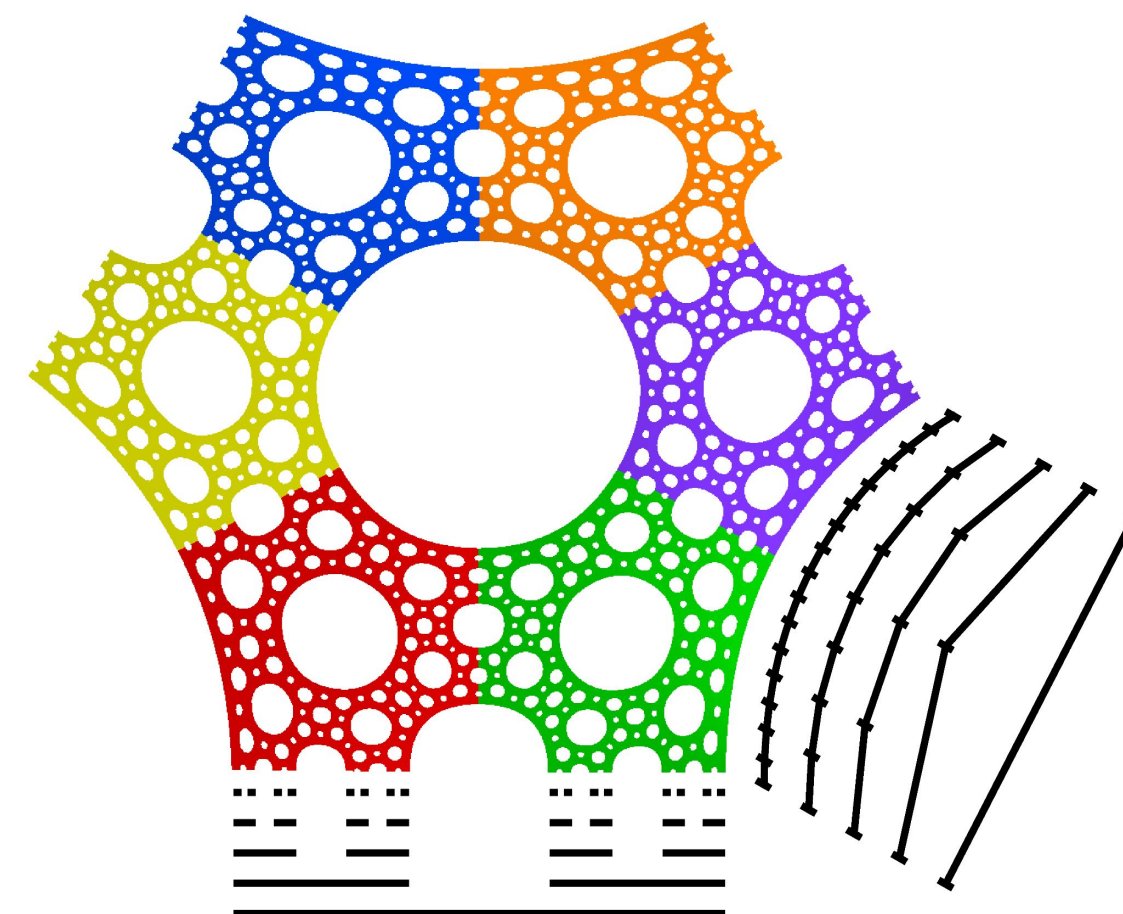
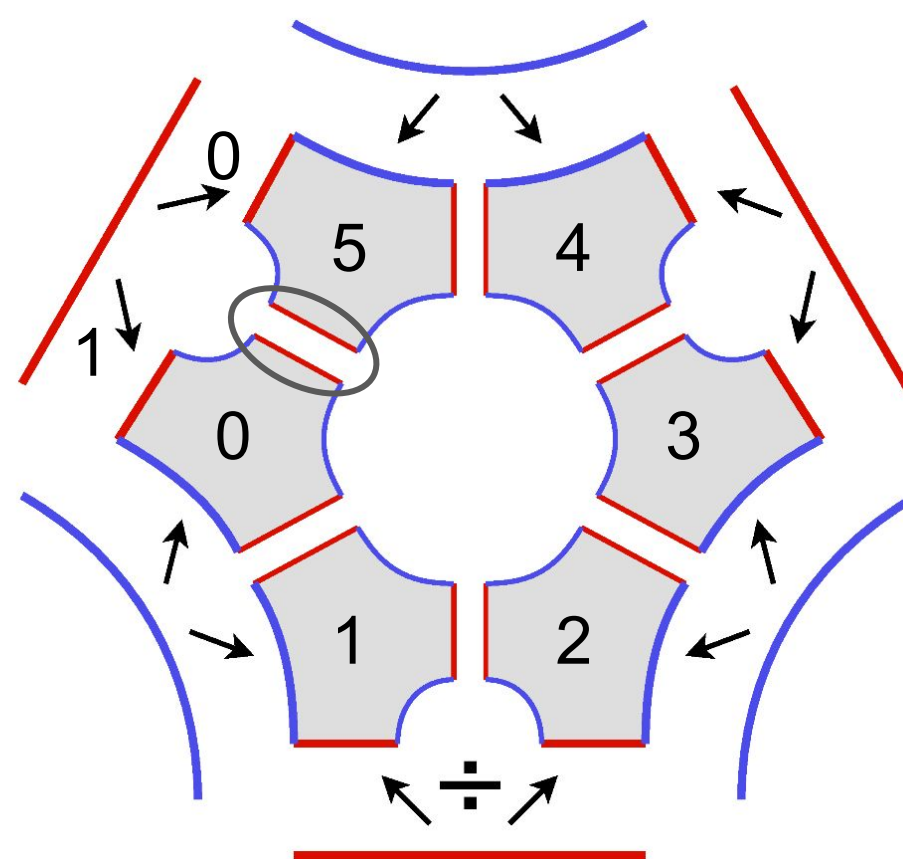
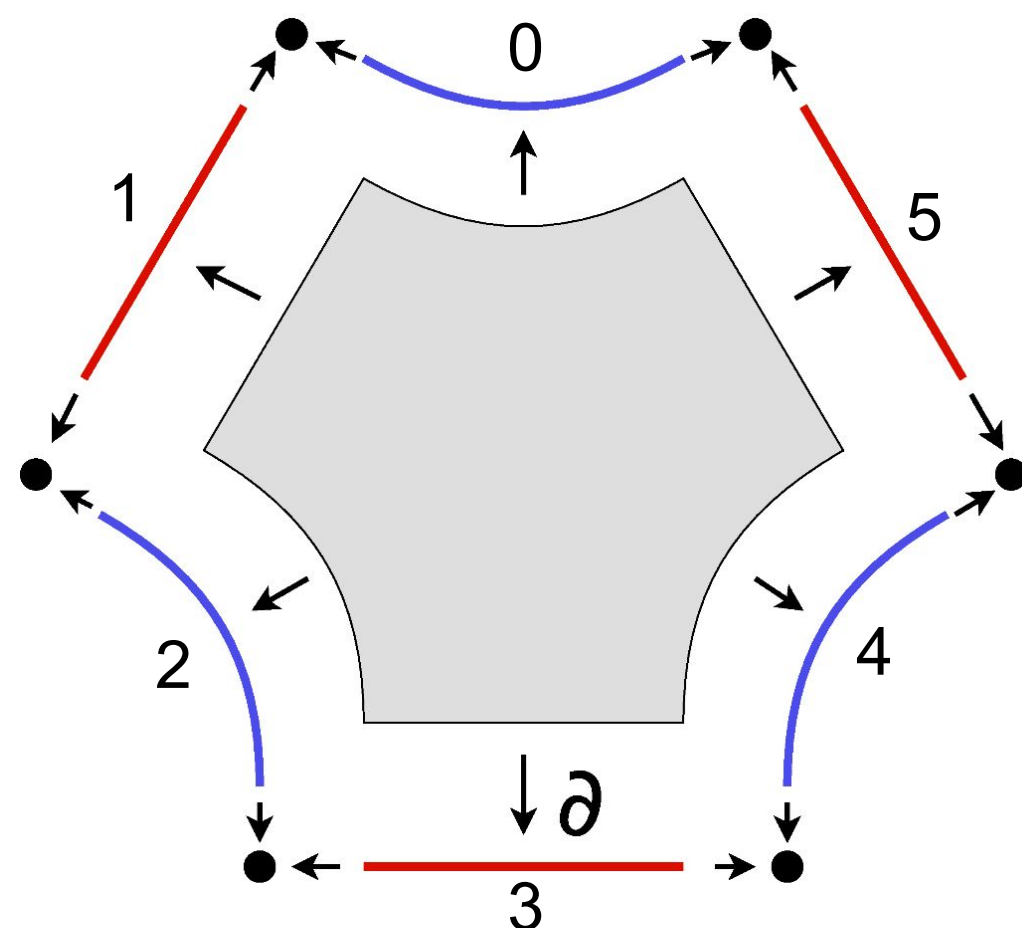
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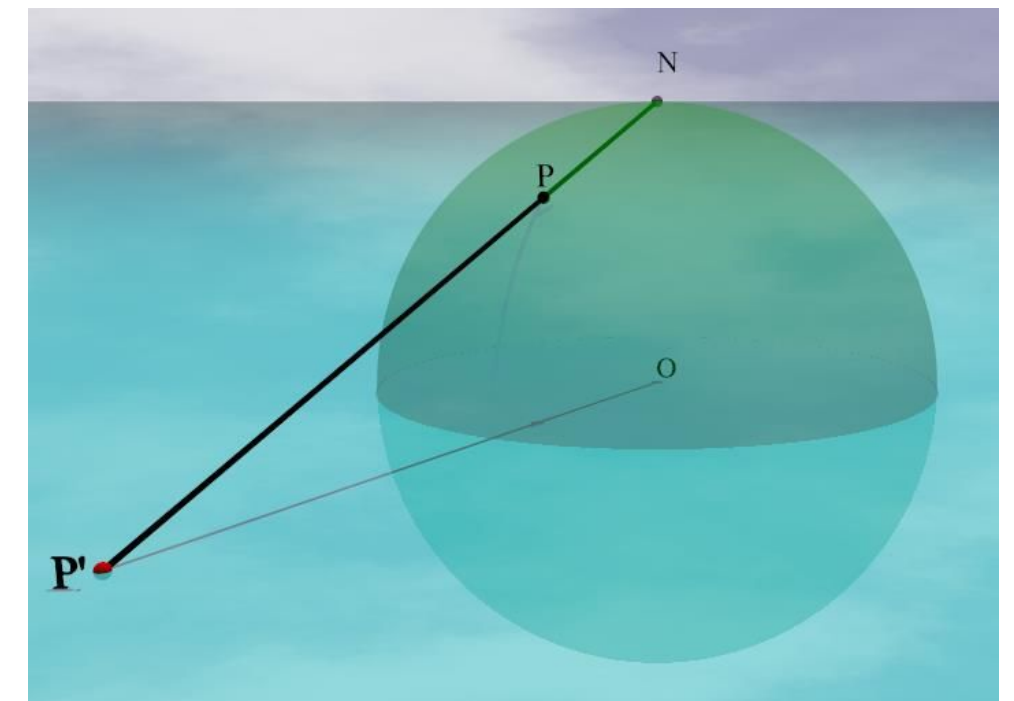
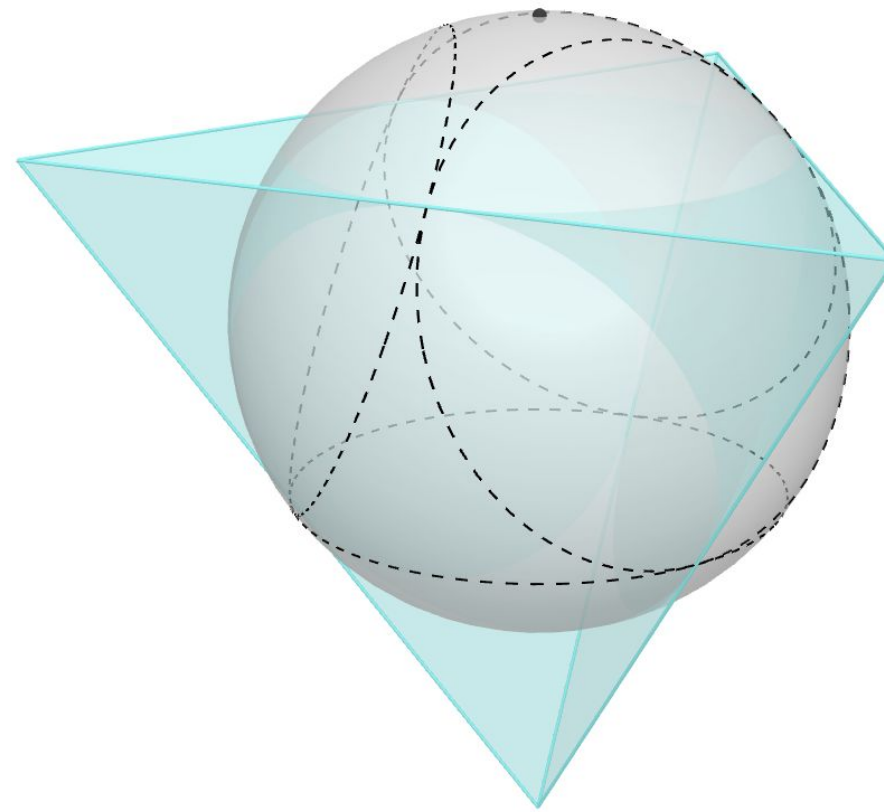
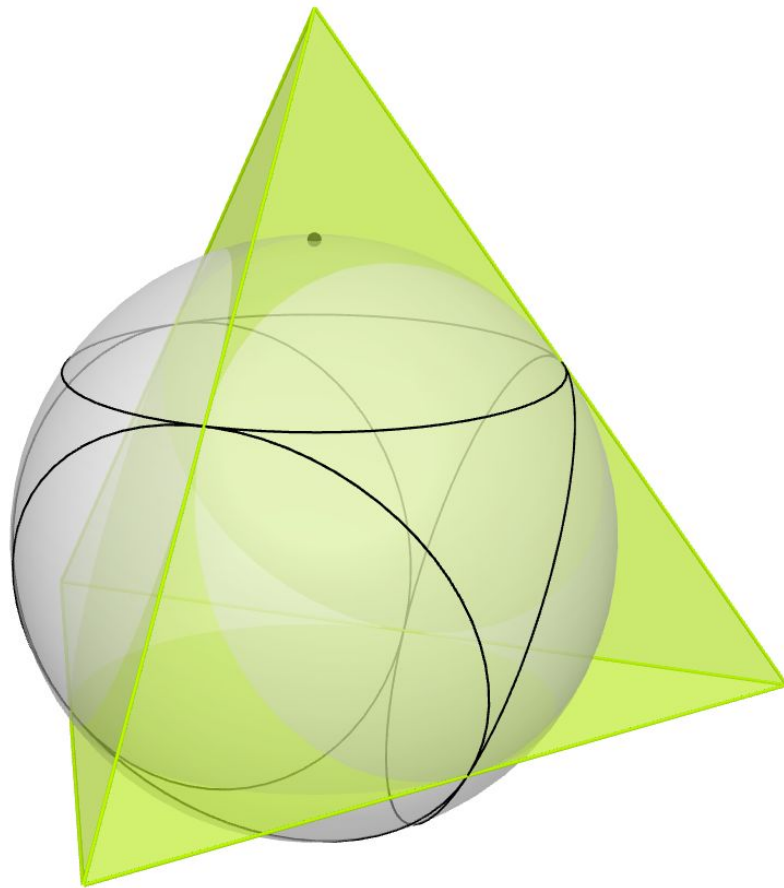


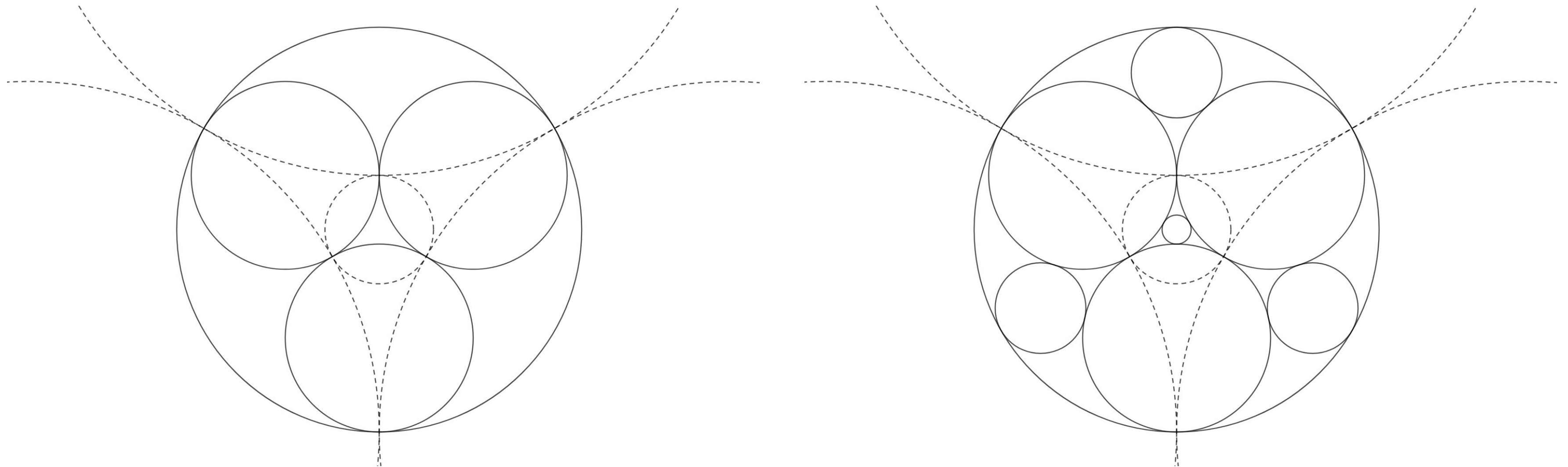


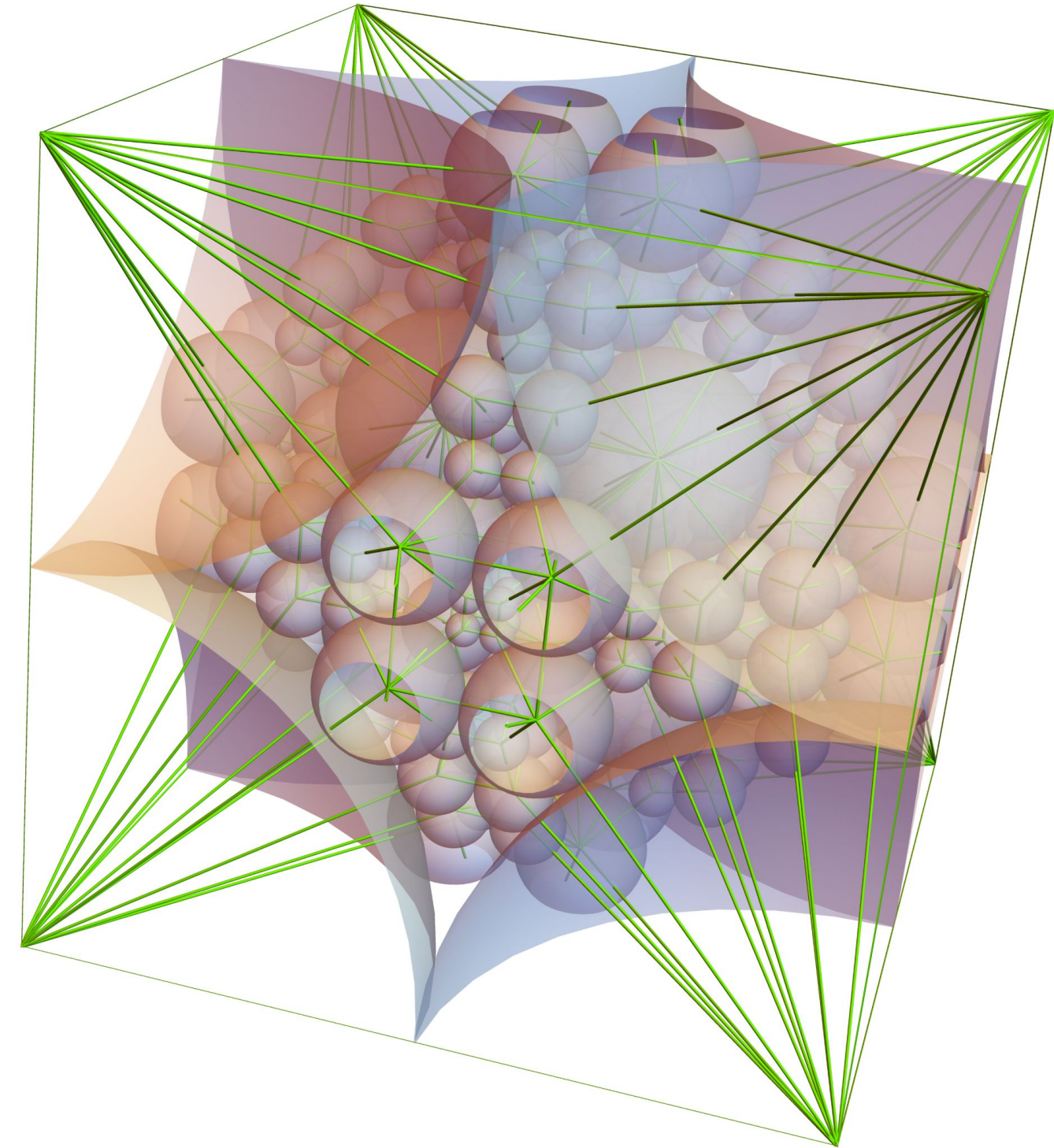
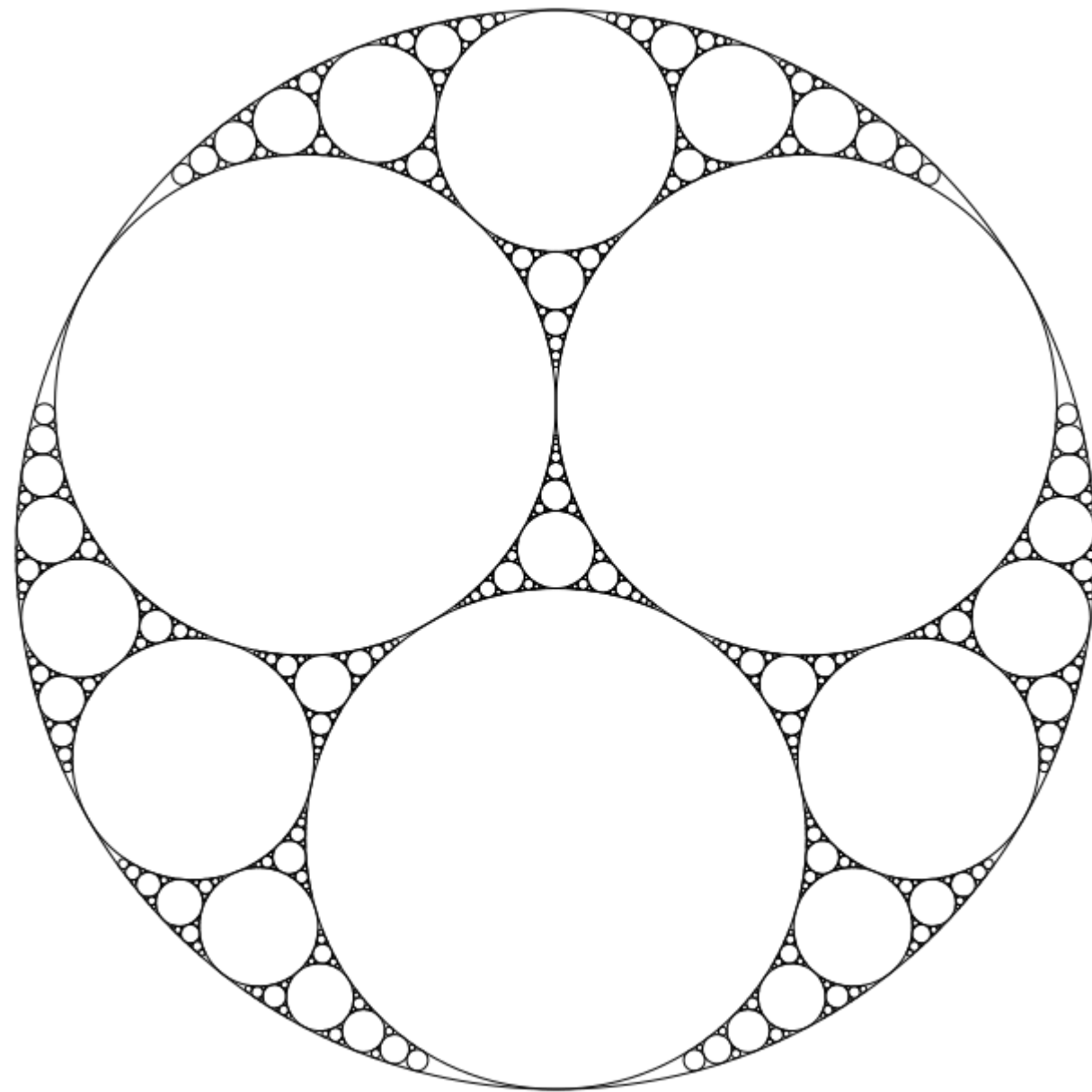
Incidence : $\text{Sub}(1) \text{ de } \text{Bord}(1) = \text{Bord}(1) \text{ de } \text{Sub}(0)$



Adjacence : $\text{Bord}(5) \text{ de } \text{Sub}(0) = \text{Bord}(3) \text{ de } \text{Sub}(5)$



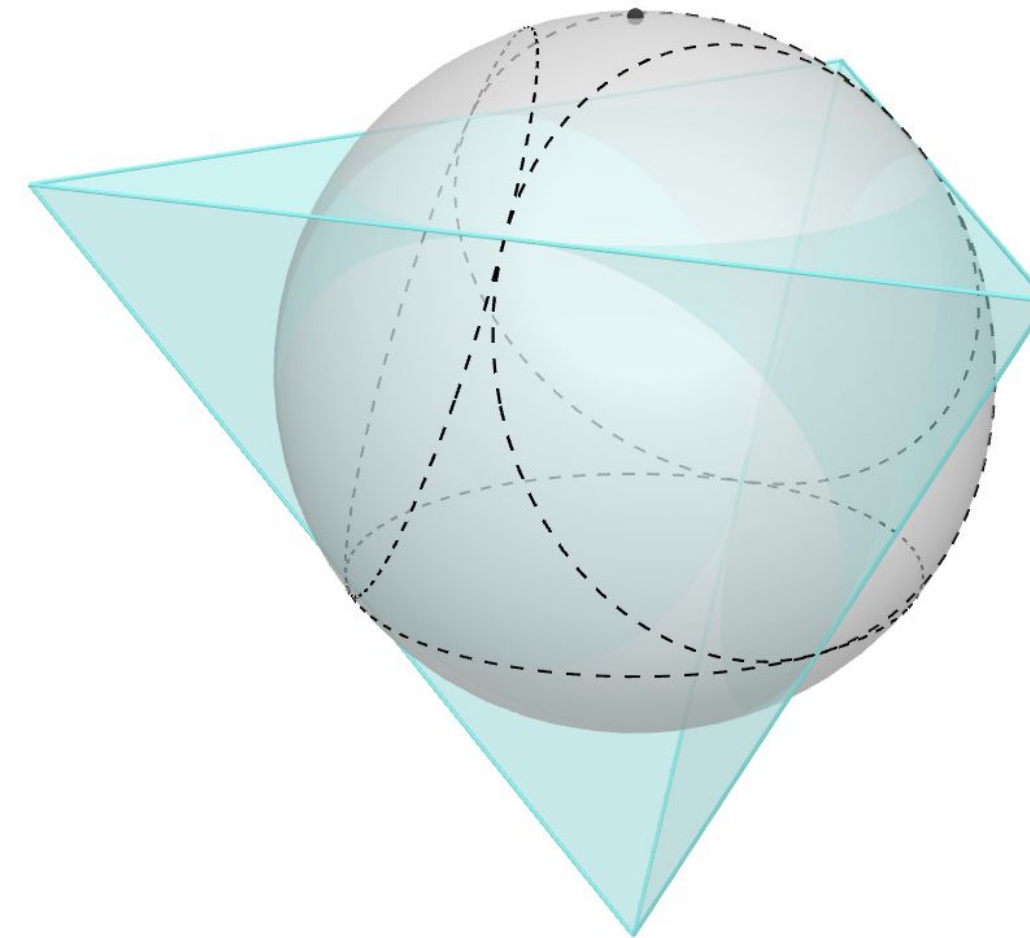
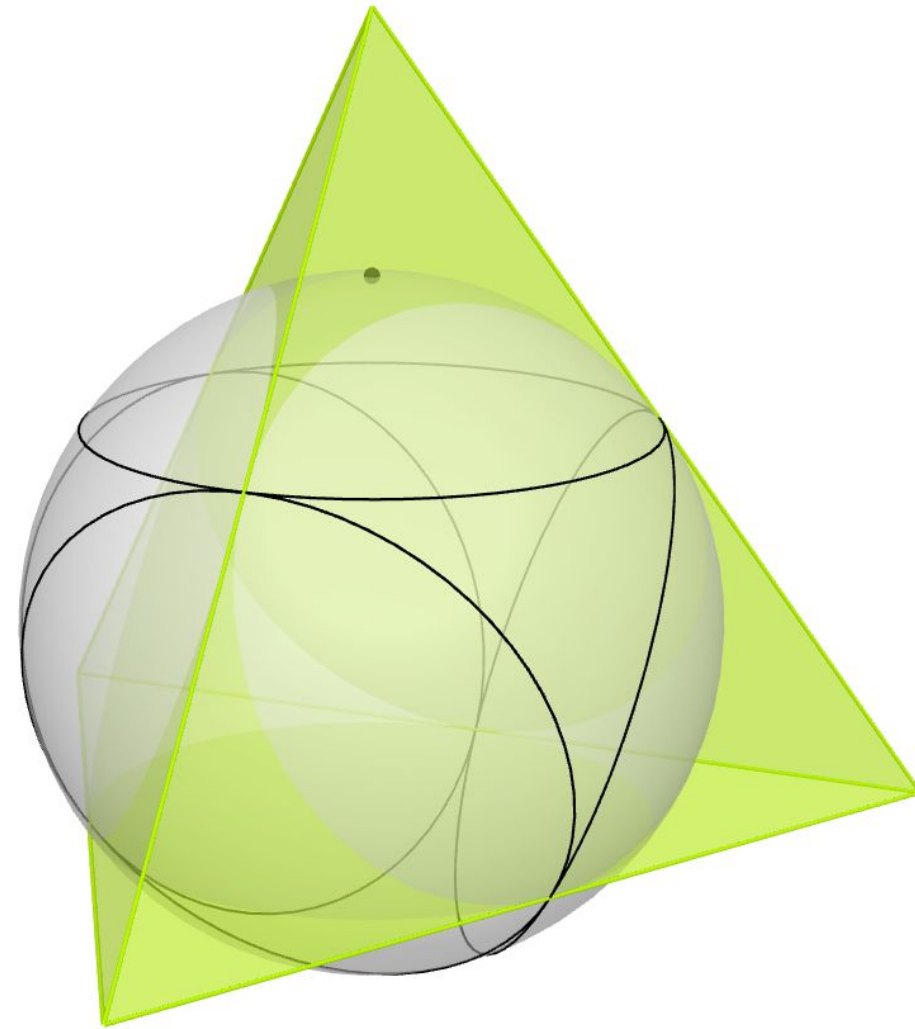


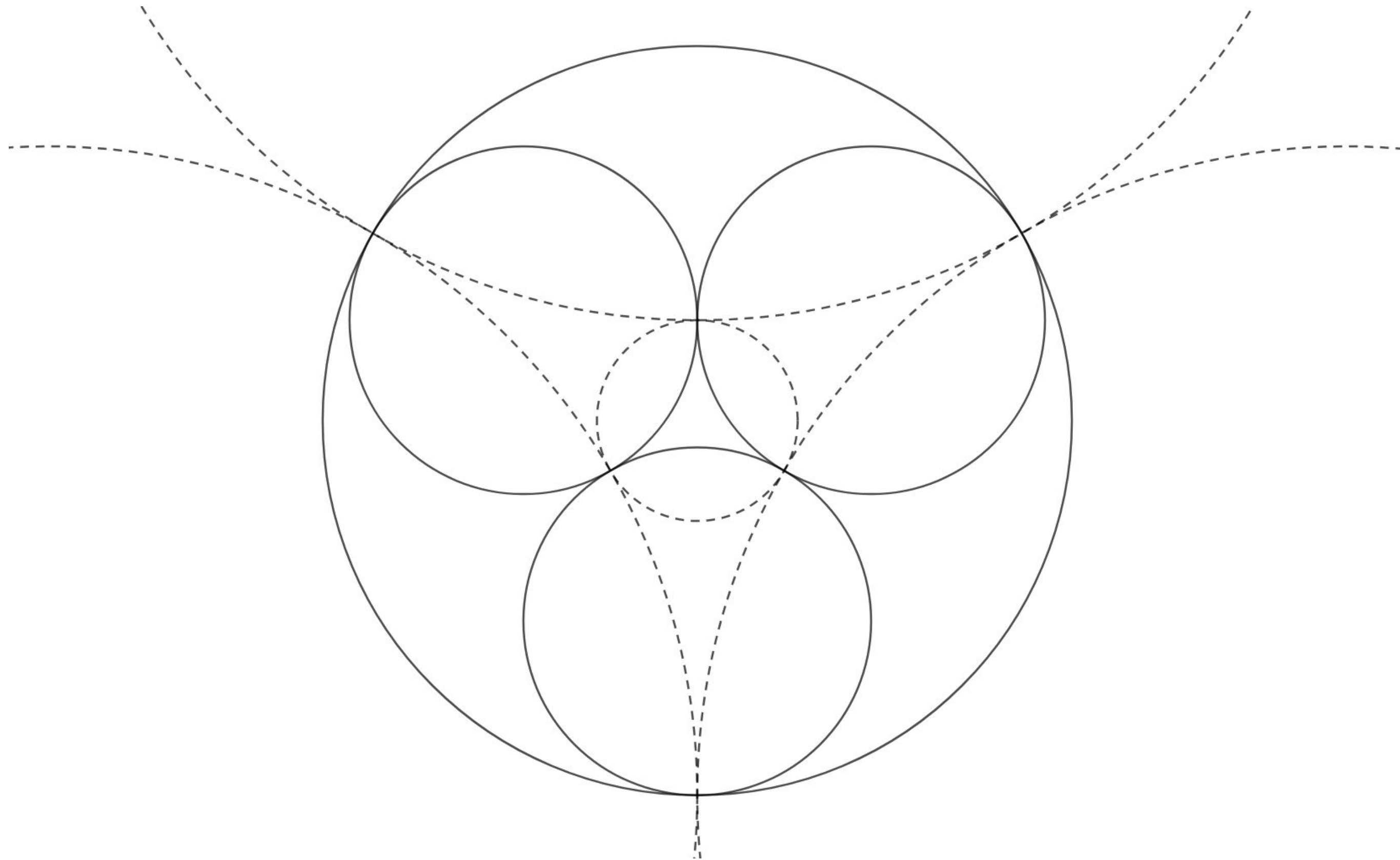


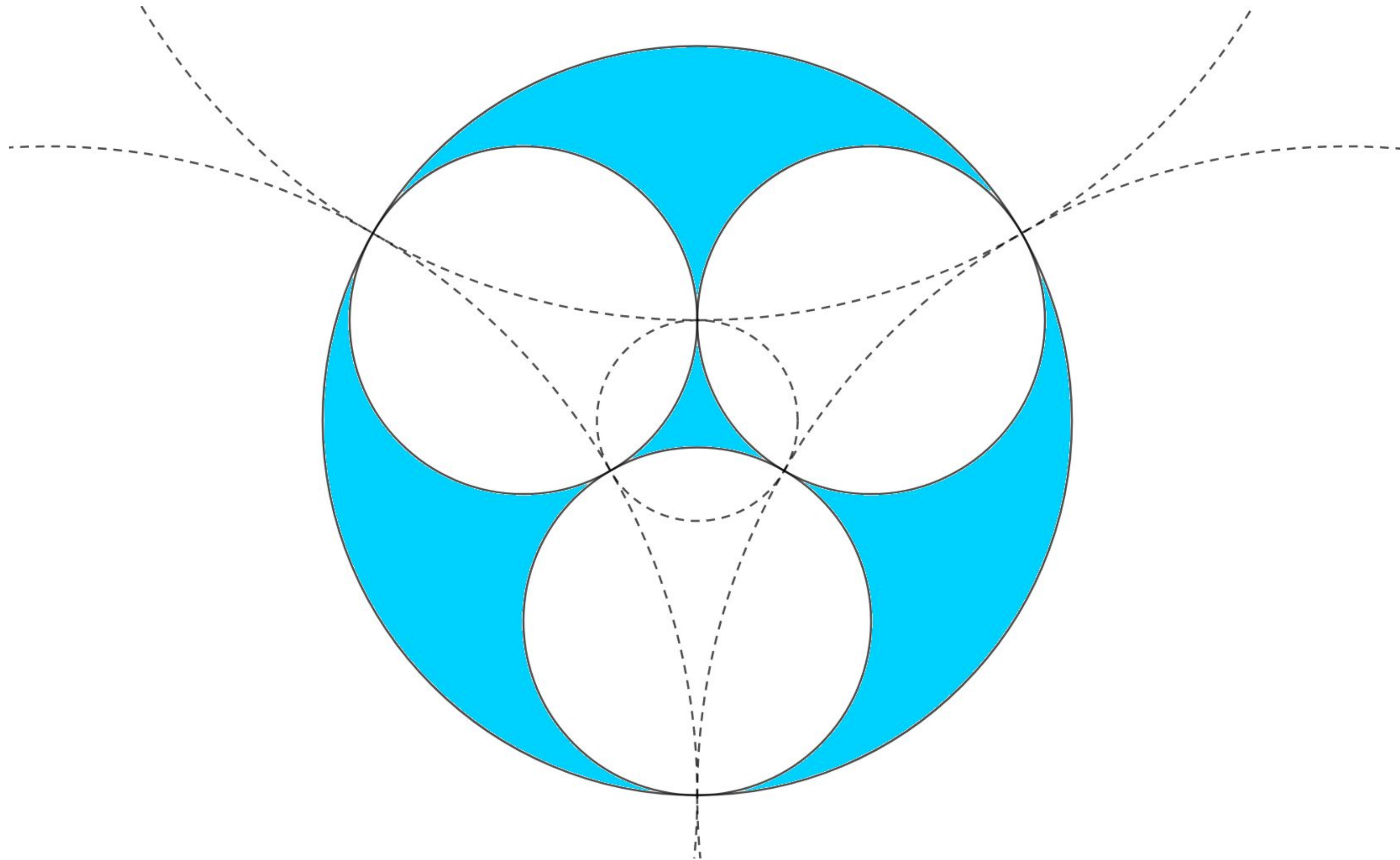
- Des empilements de sphères aux BC-IFS
- Extraction de règles de construction
- Conclusion

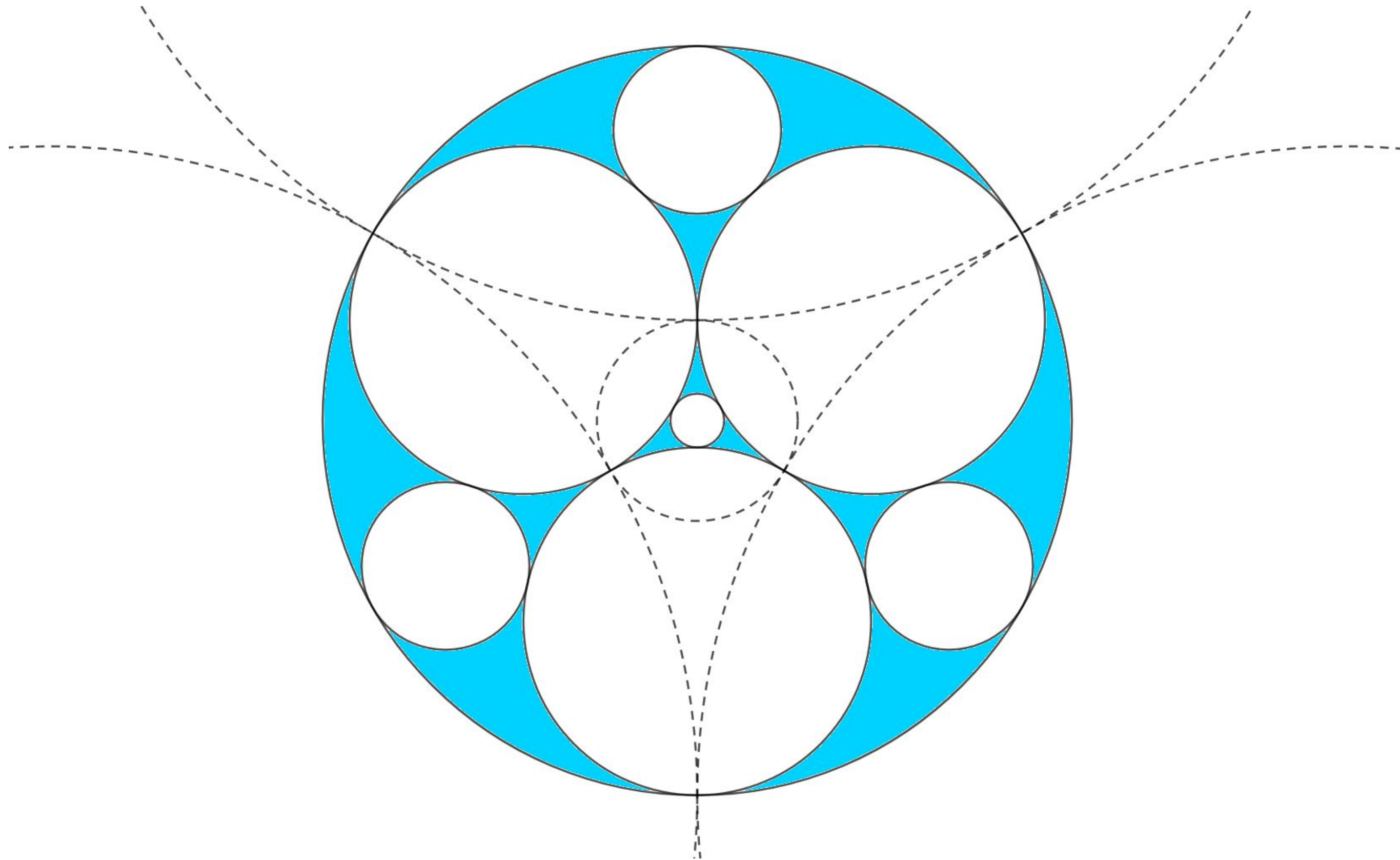
Des empilements de sphères aux BC-IFS

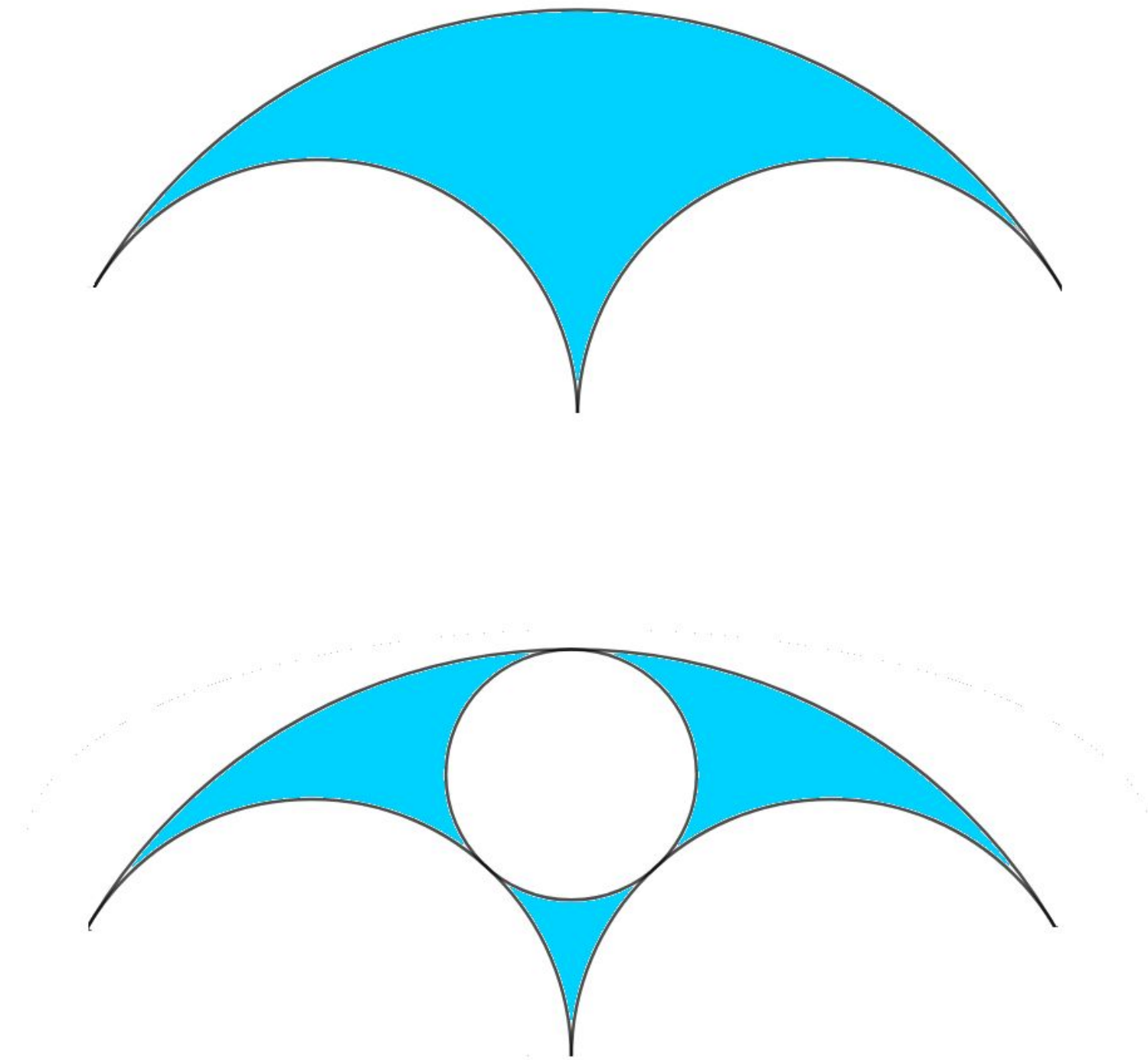


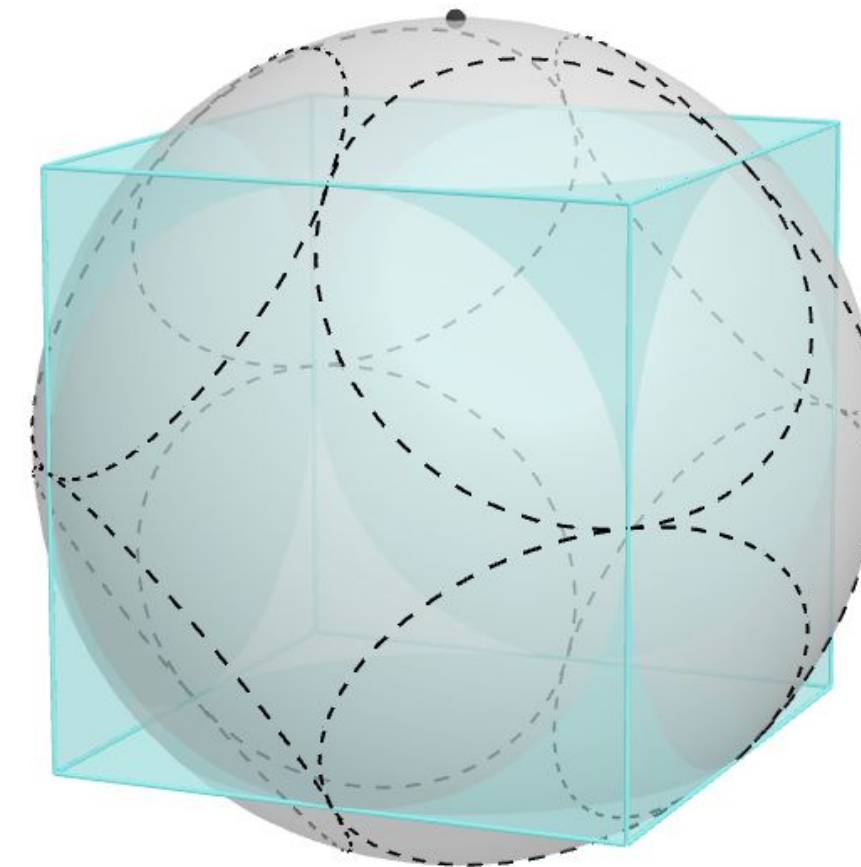
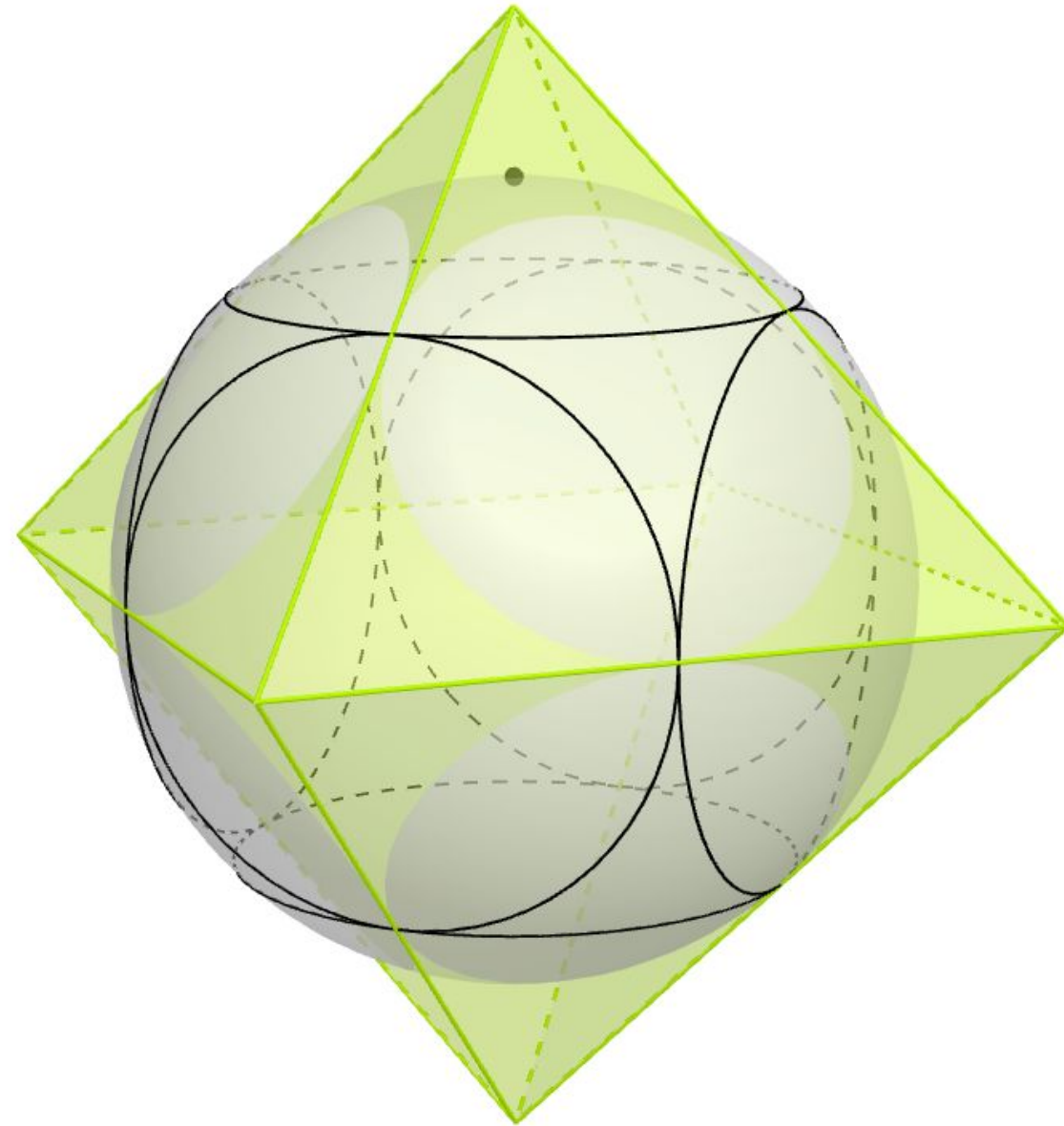


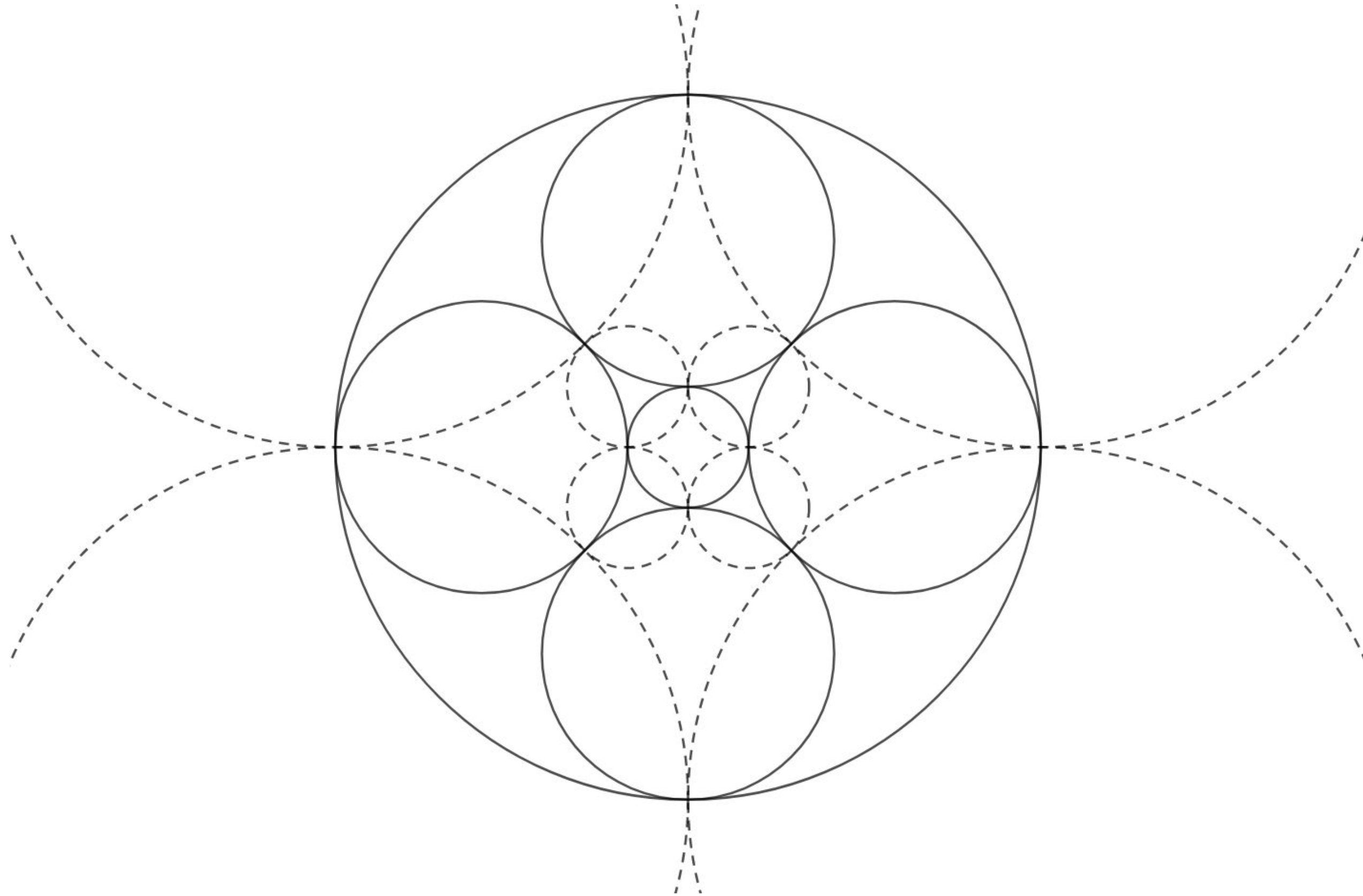


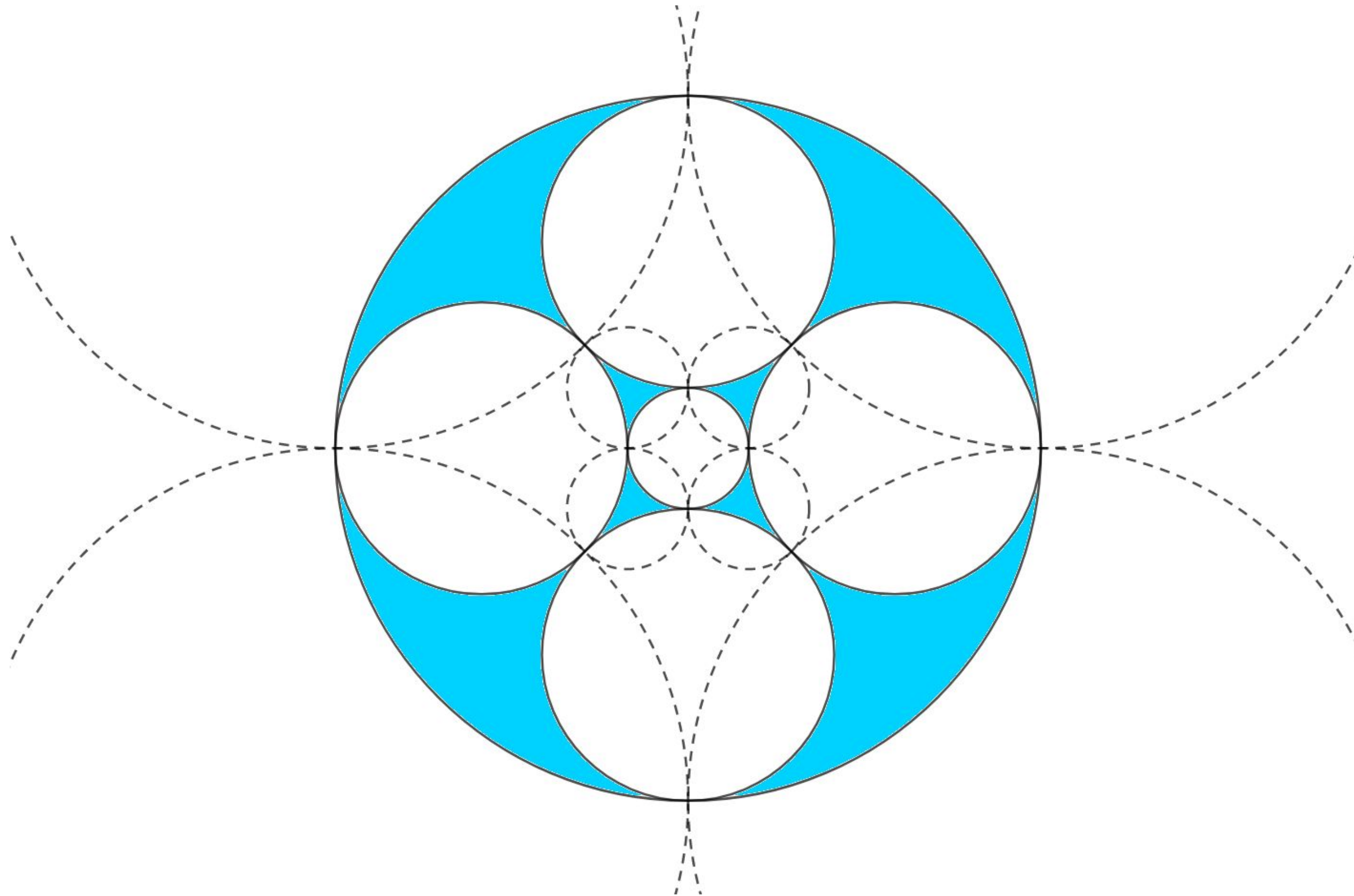


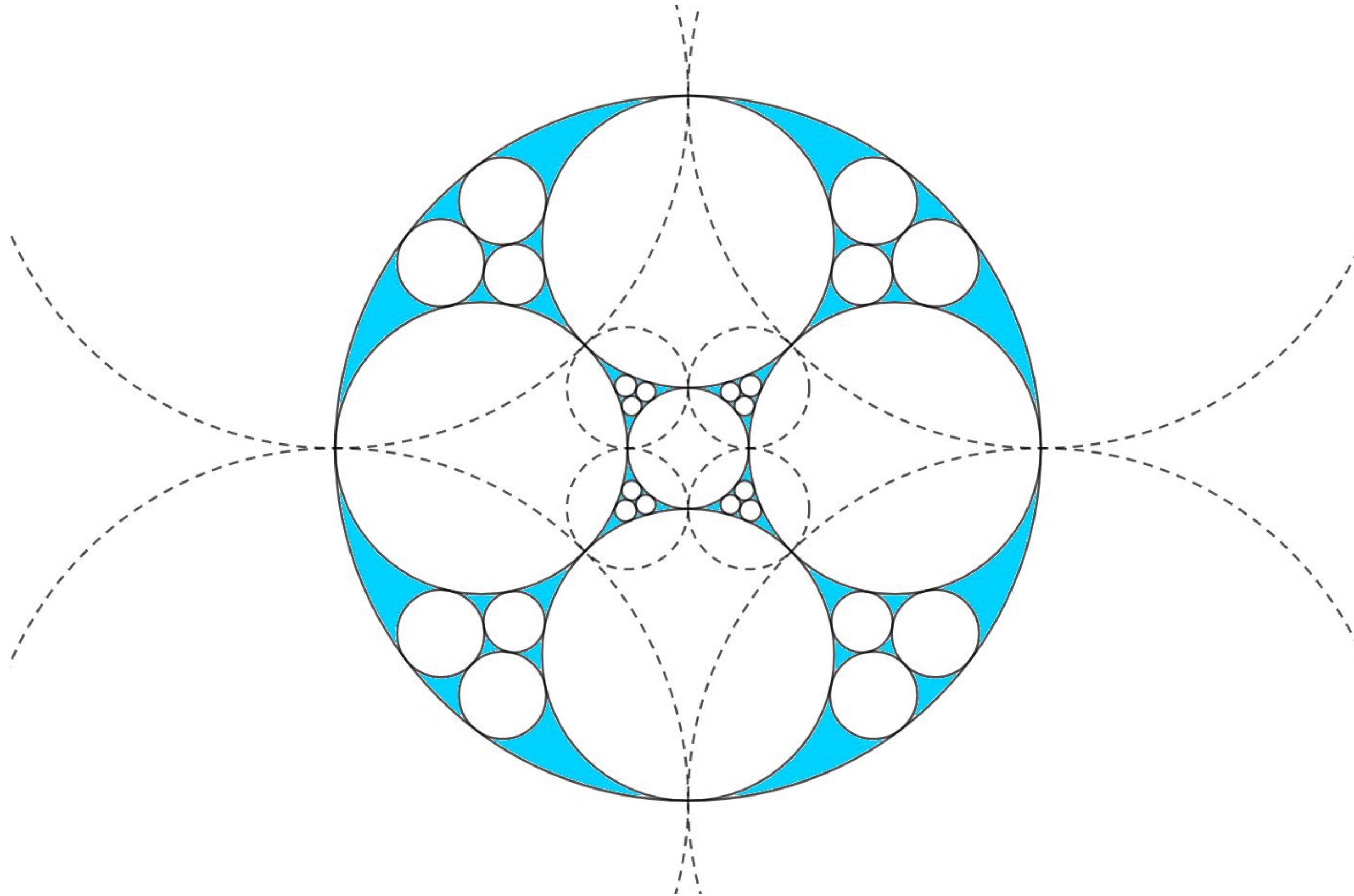


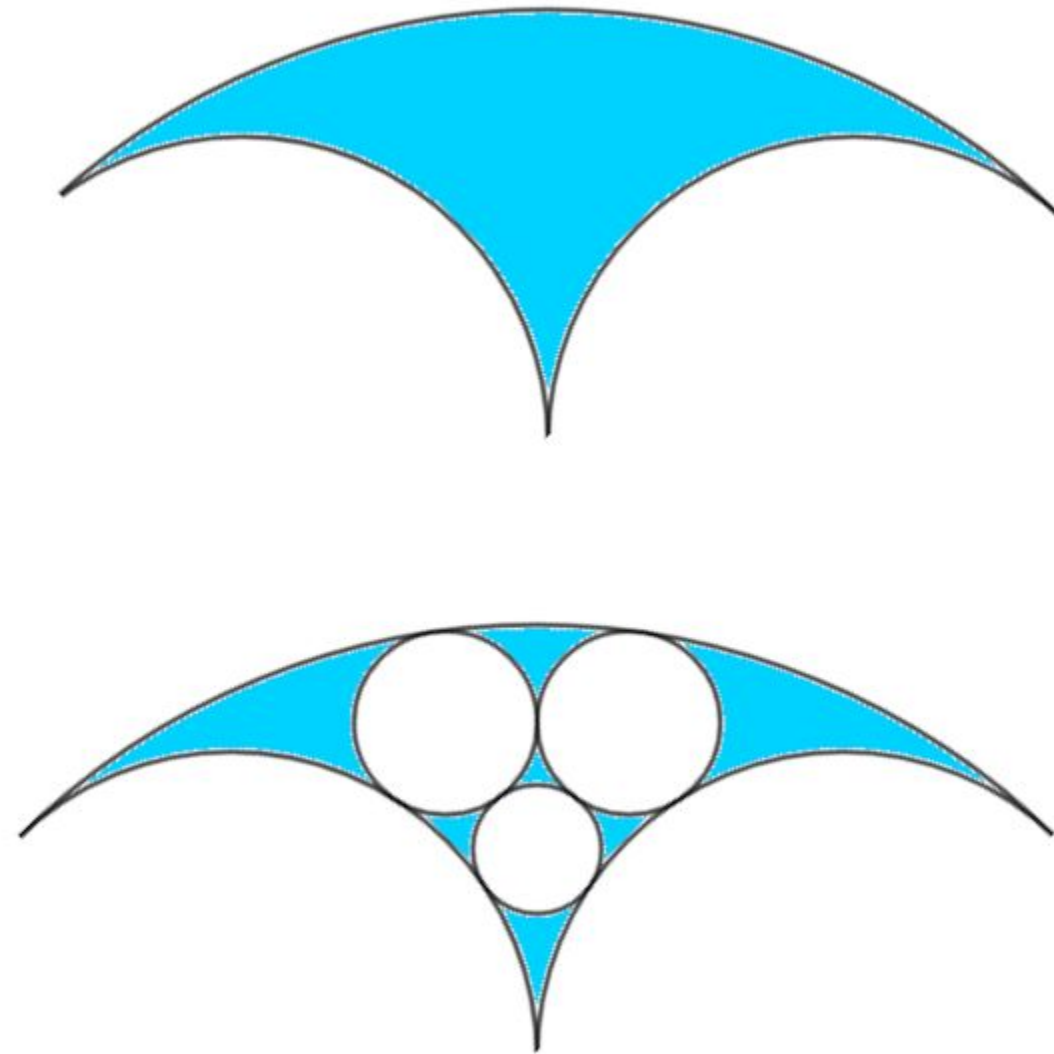












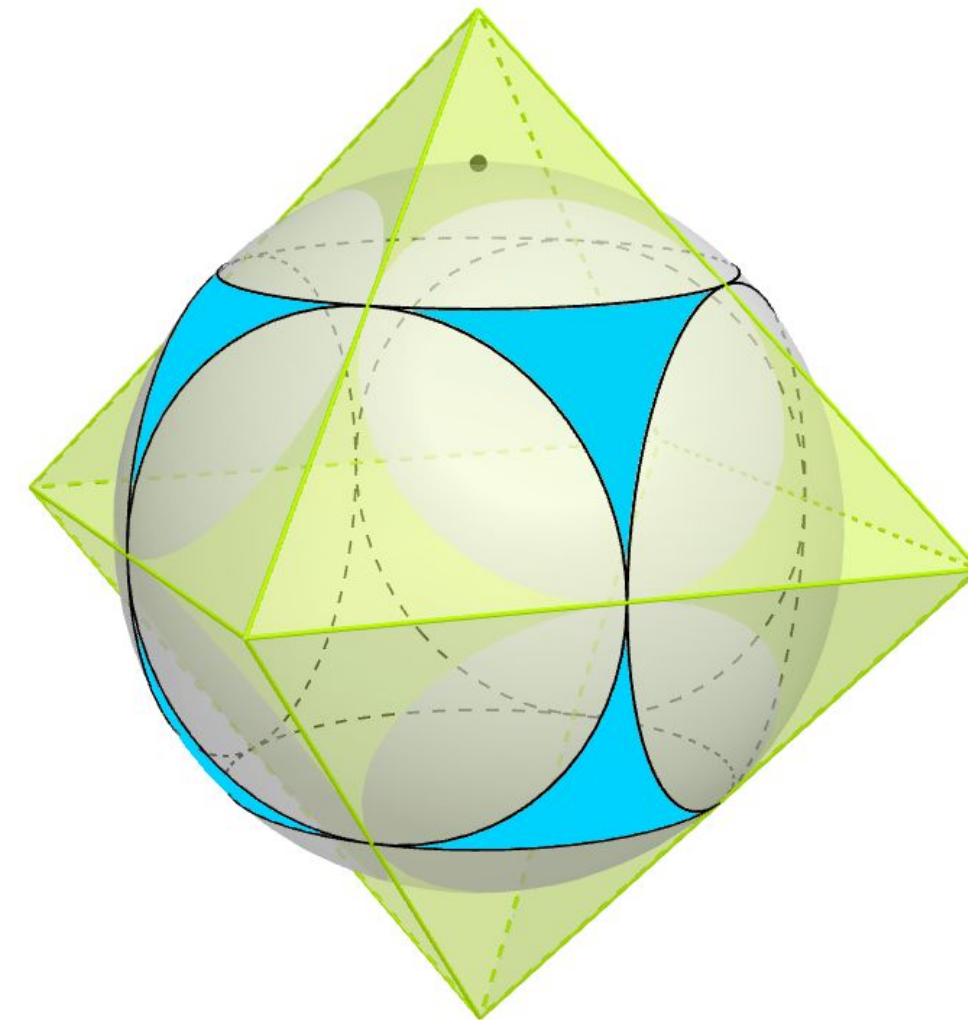
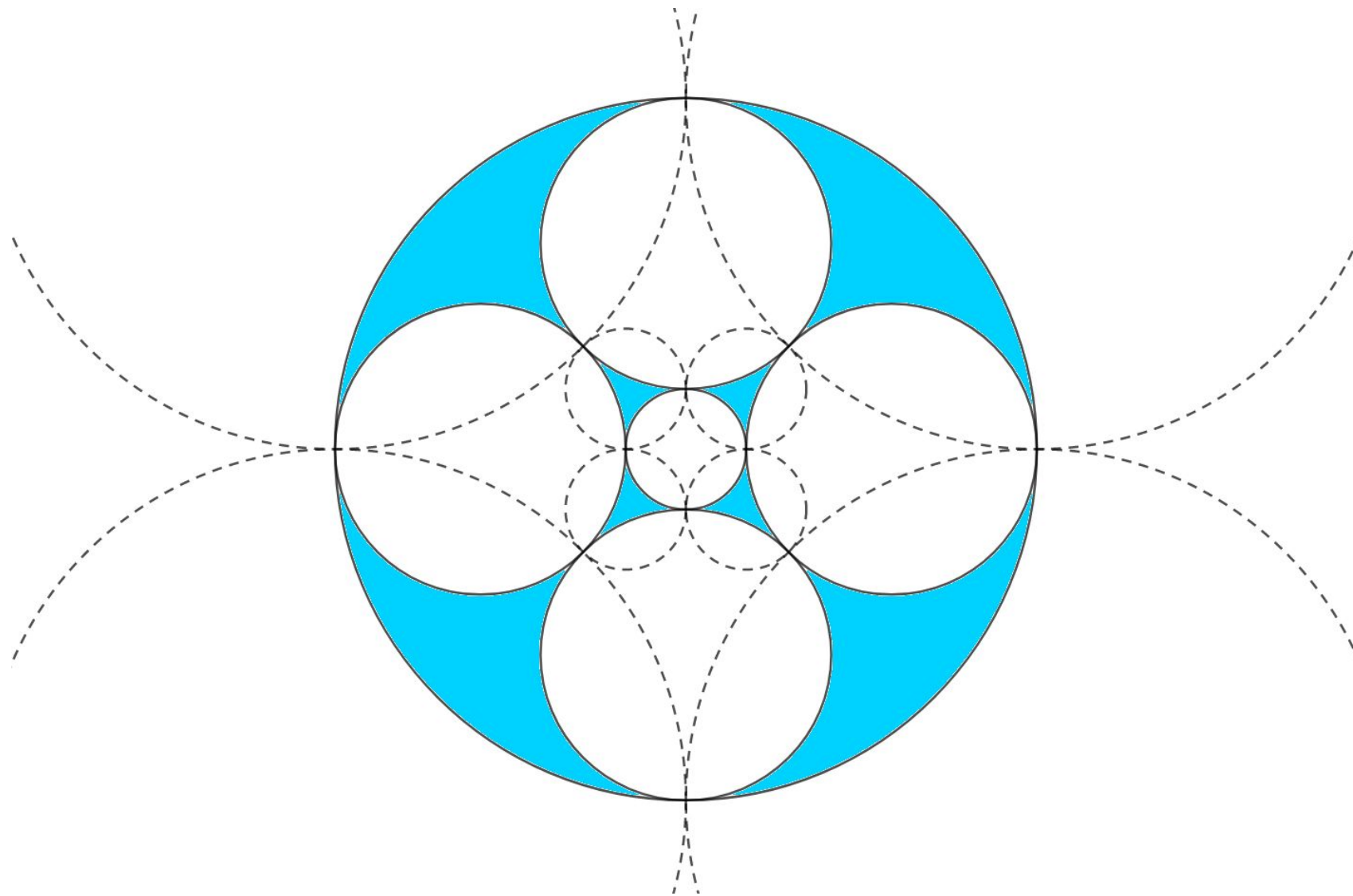
Extraction de règles de construction



- Combien de faces ?
- Comment sont-elles agencées ?
- Combien de bords par face ?
- Comment se subdivisent-elles ?
- Quelles contraintes d'adjacence ?
- Comment se subdivisent les bords ?
- Quelles contraintes d'incidence ?

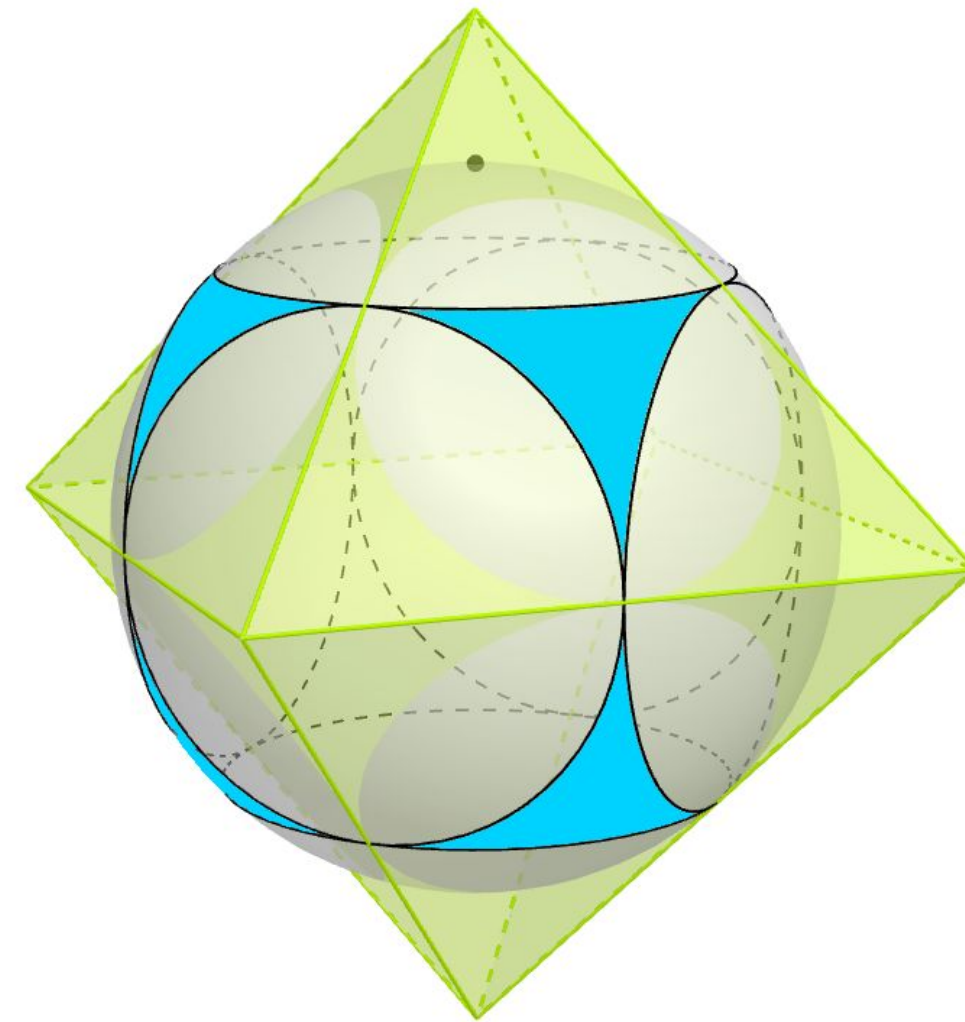
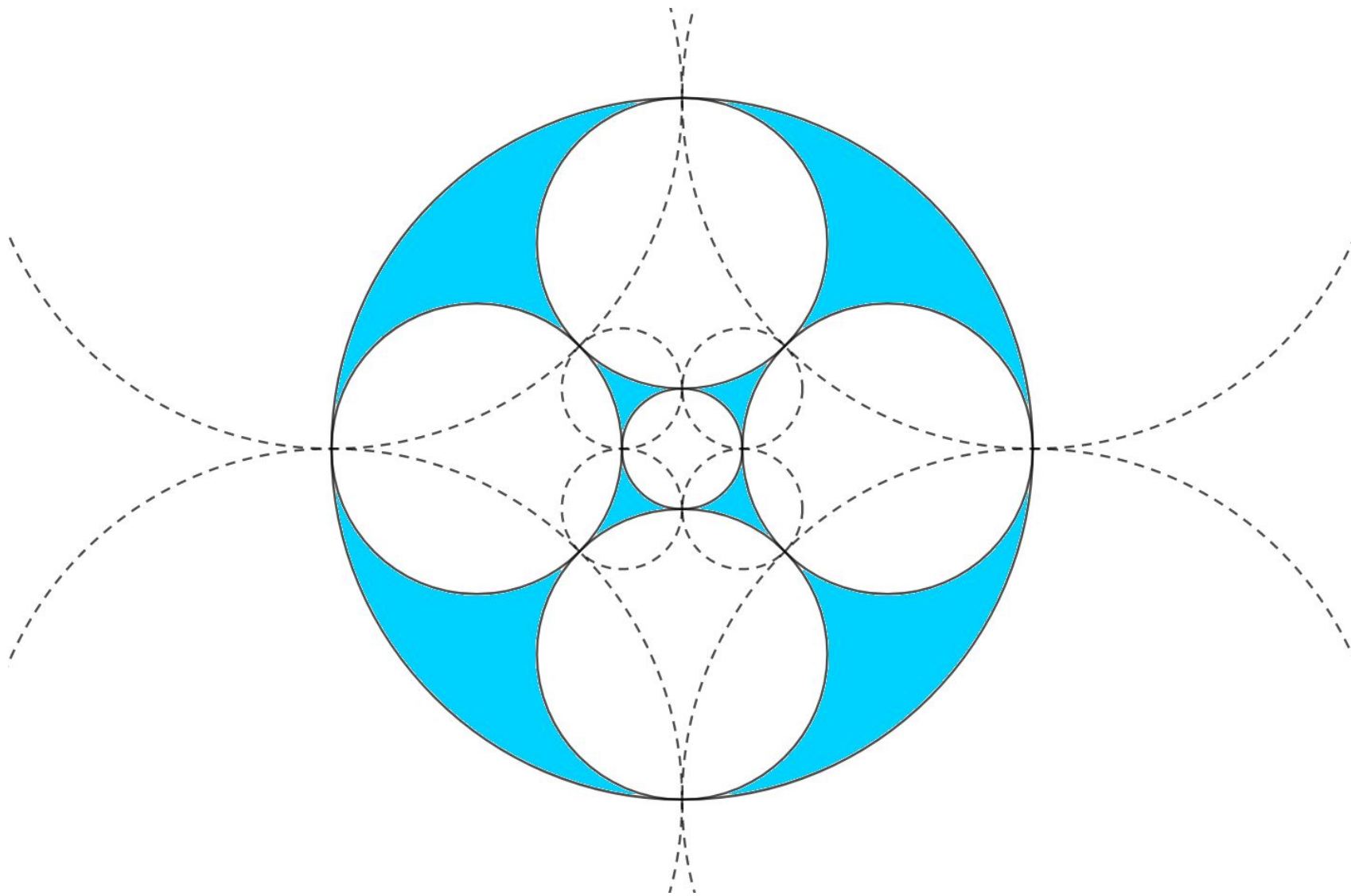
Combien de faces ?

Une face fractale par face du polyèdre



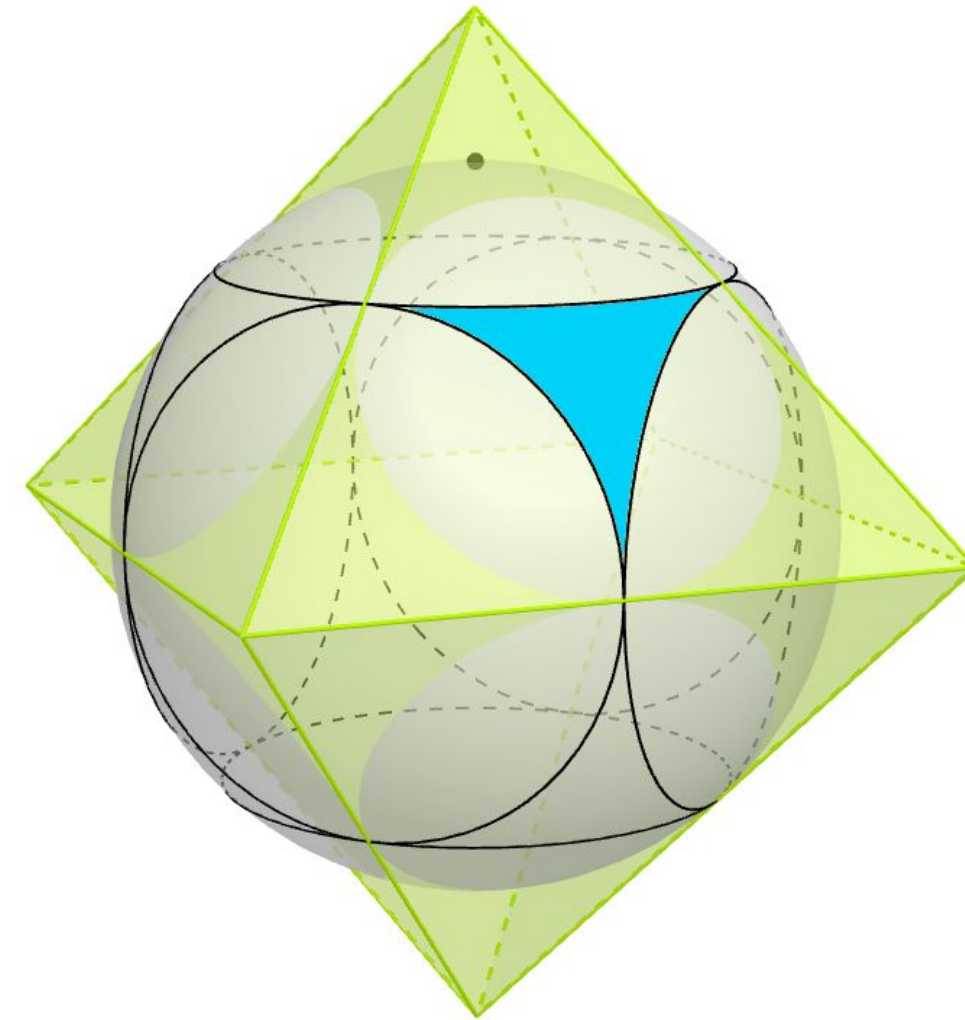
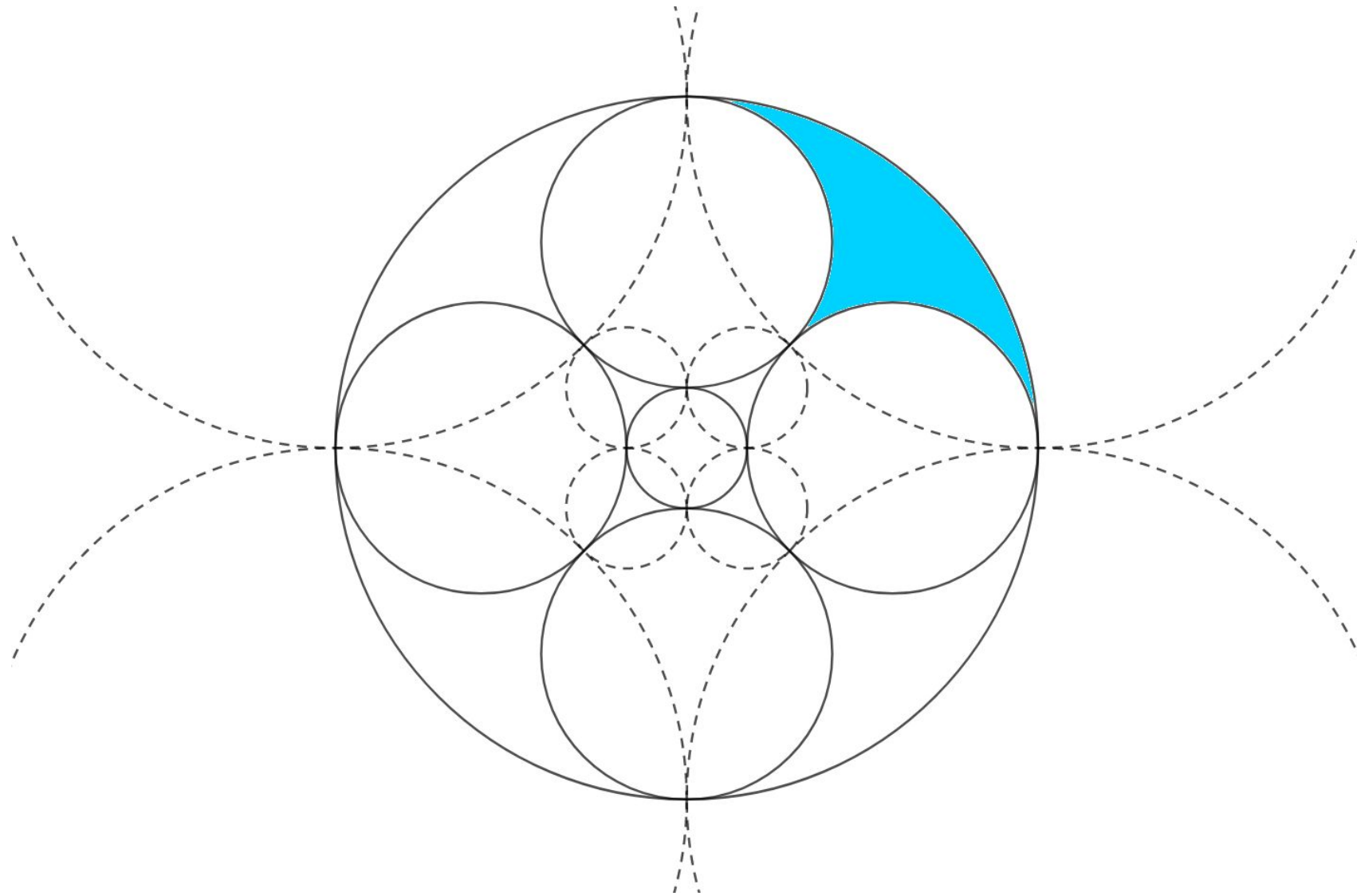
Comment sont-elles agencées ?

Adjacences des faces fractales = adjacences des faces du polyèdre



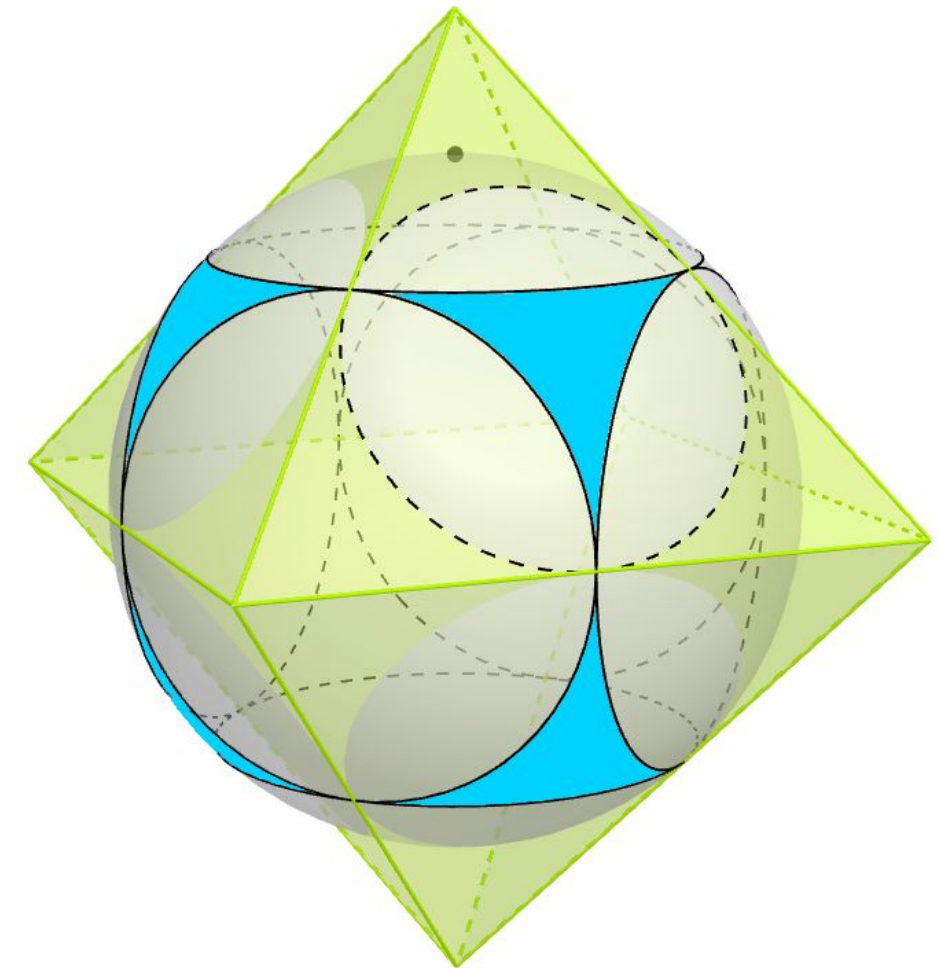
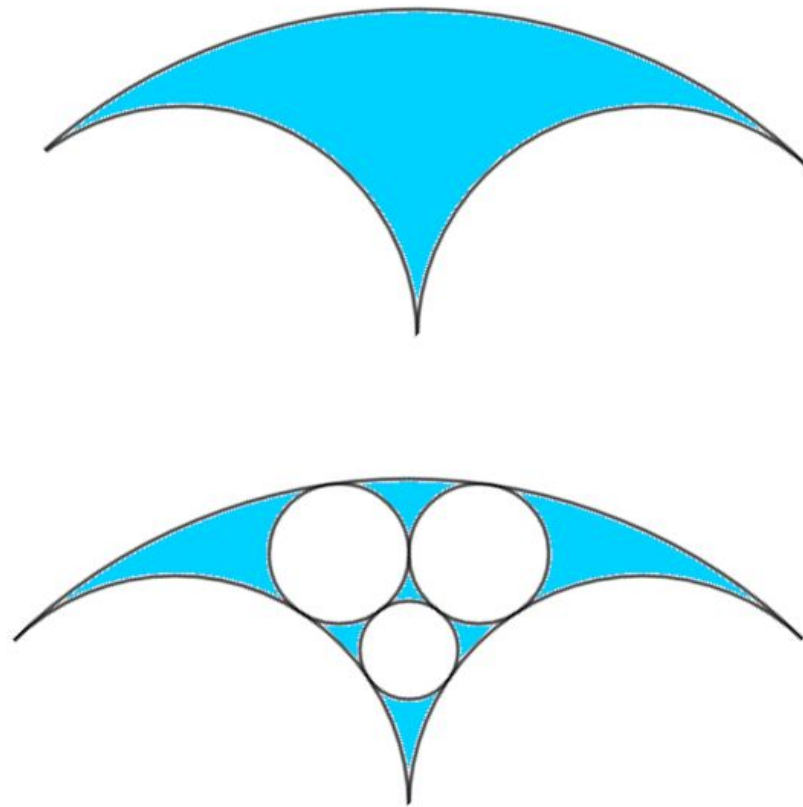
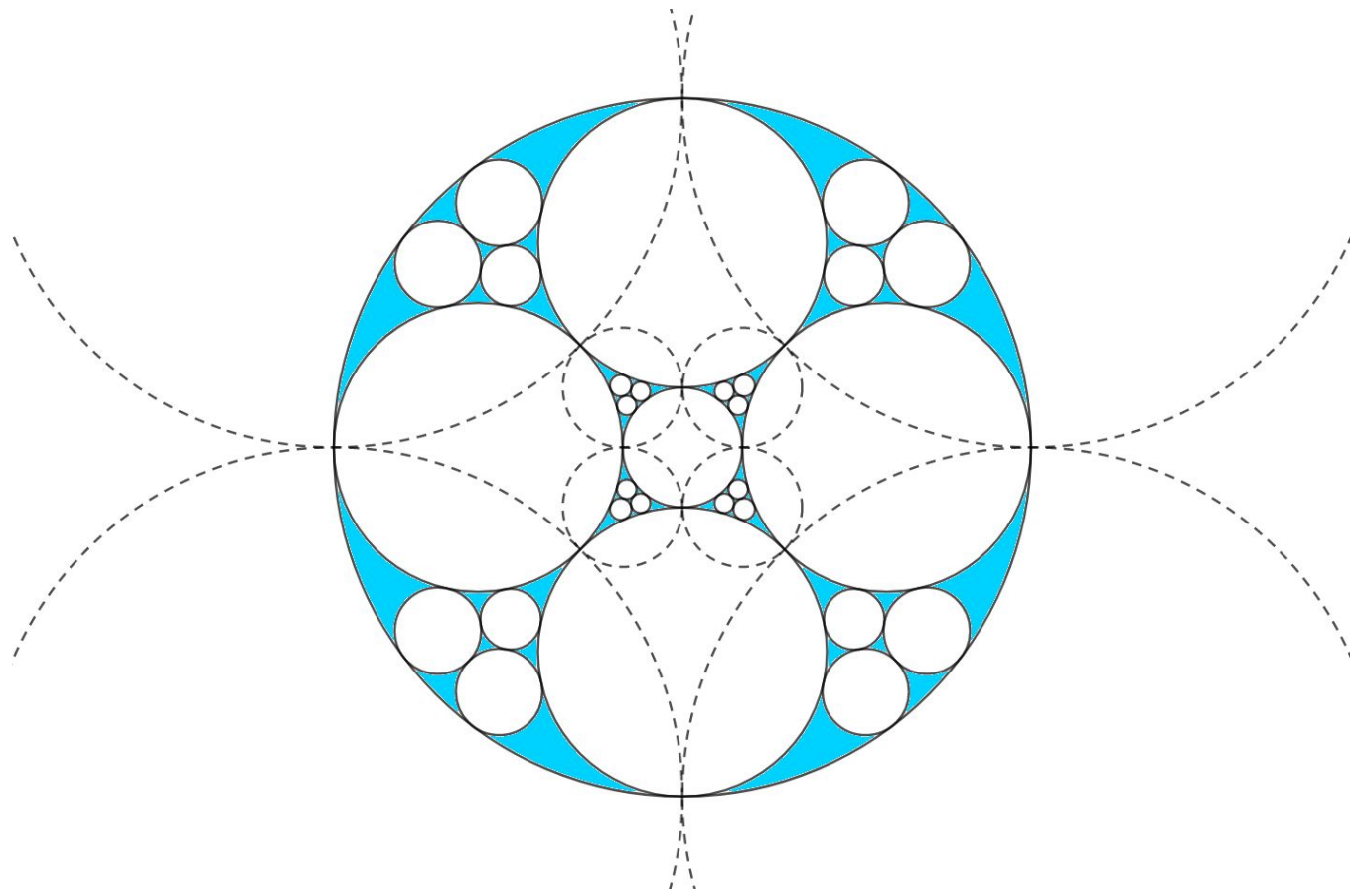
Combien de bords par face ?

Nombre de bords = nombre de sommets



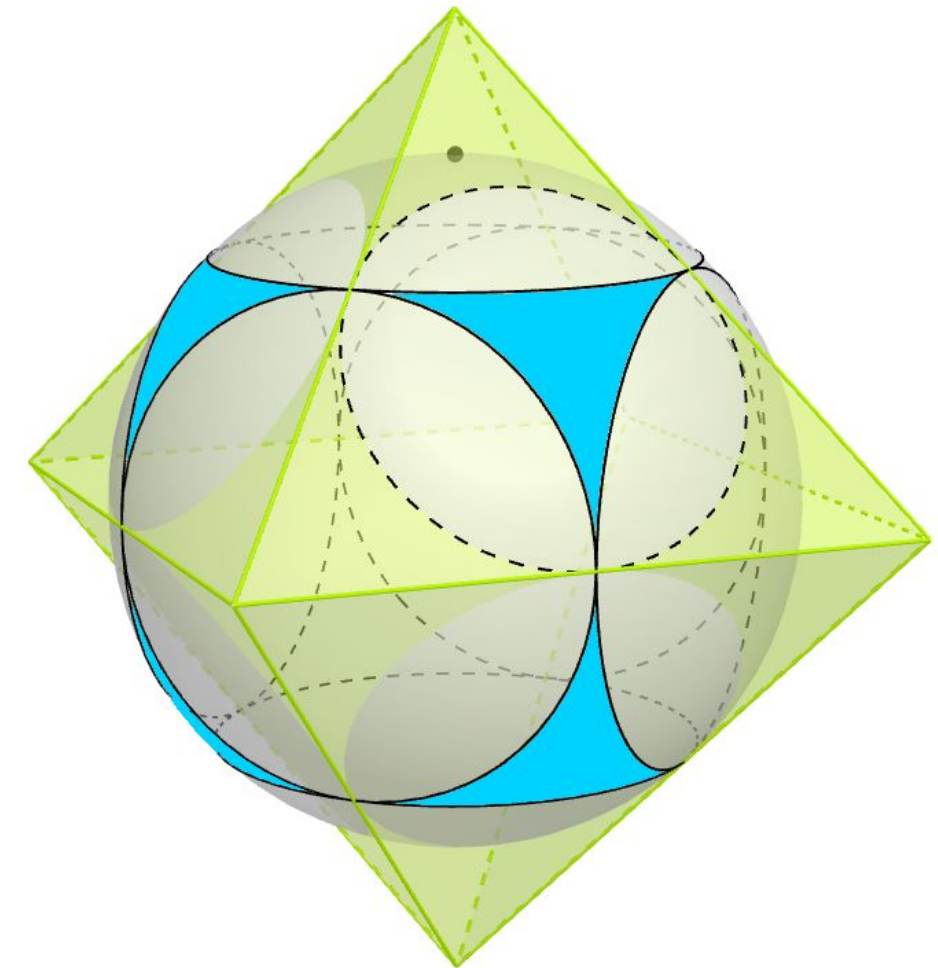
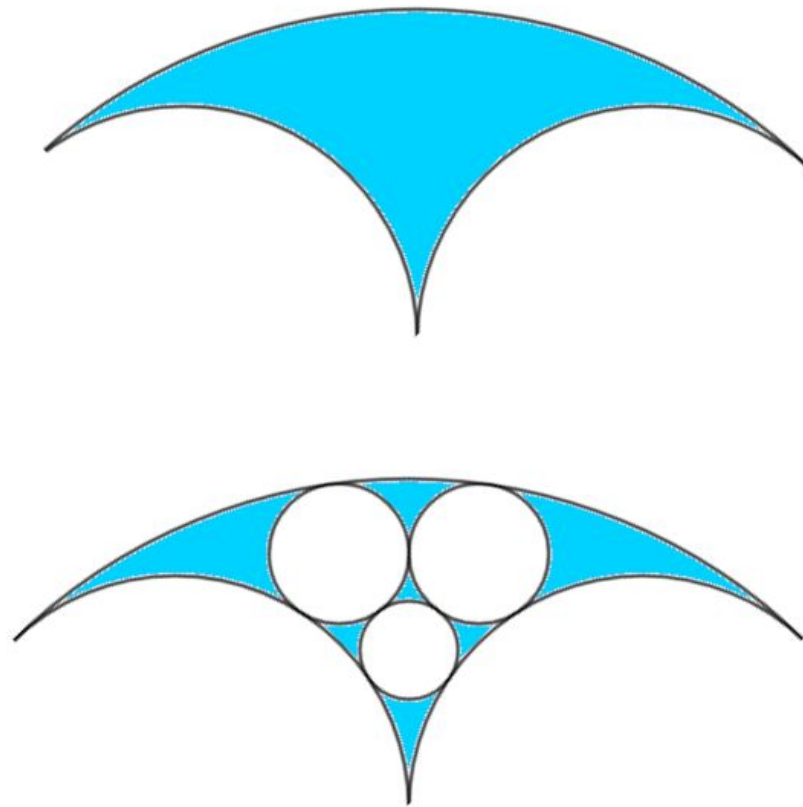
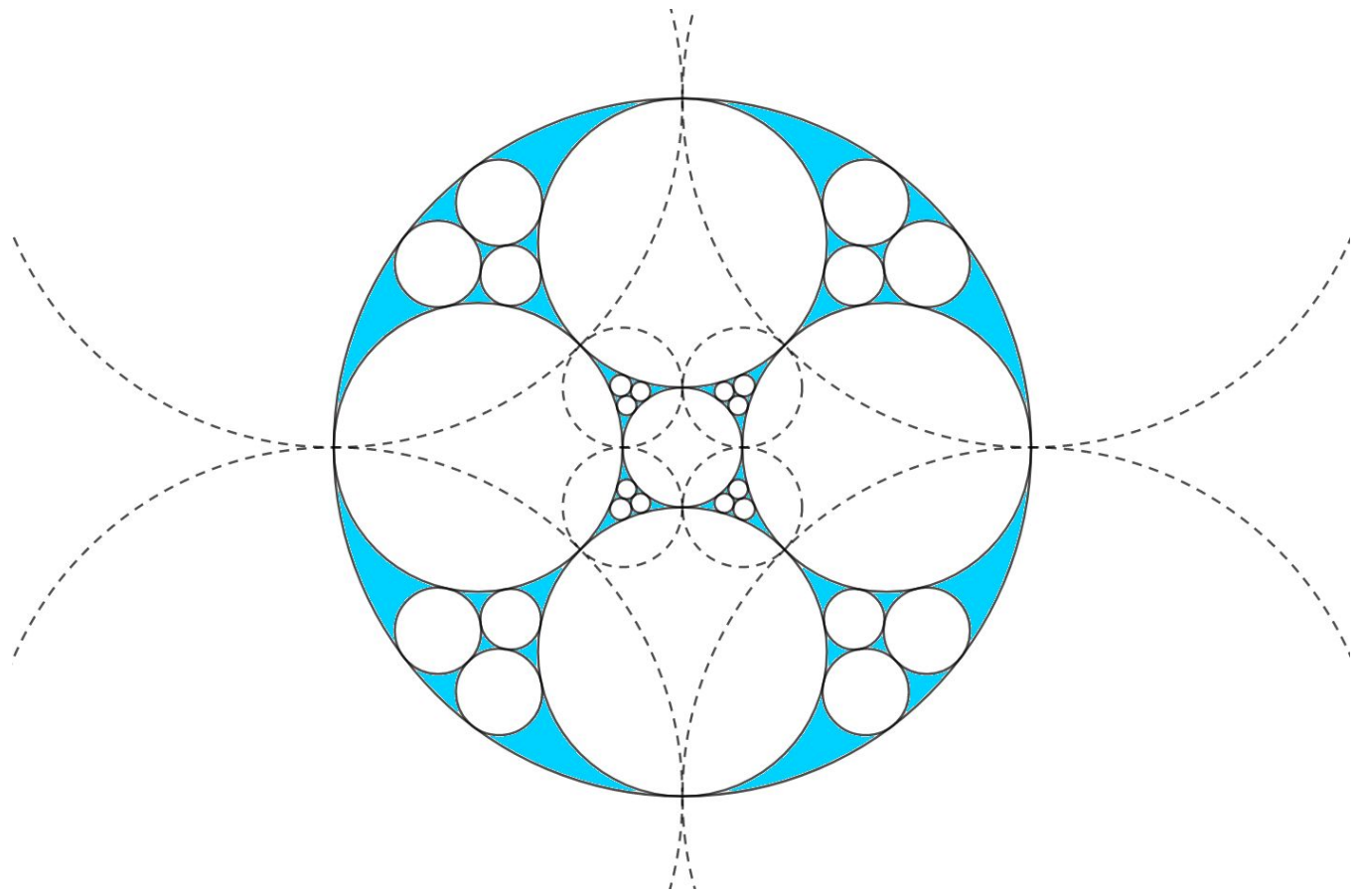
Comment se subdivisent-elles ?

Nombre de subdivisions = nombre de faces du polyèdre - 1



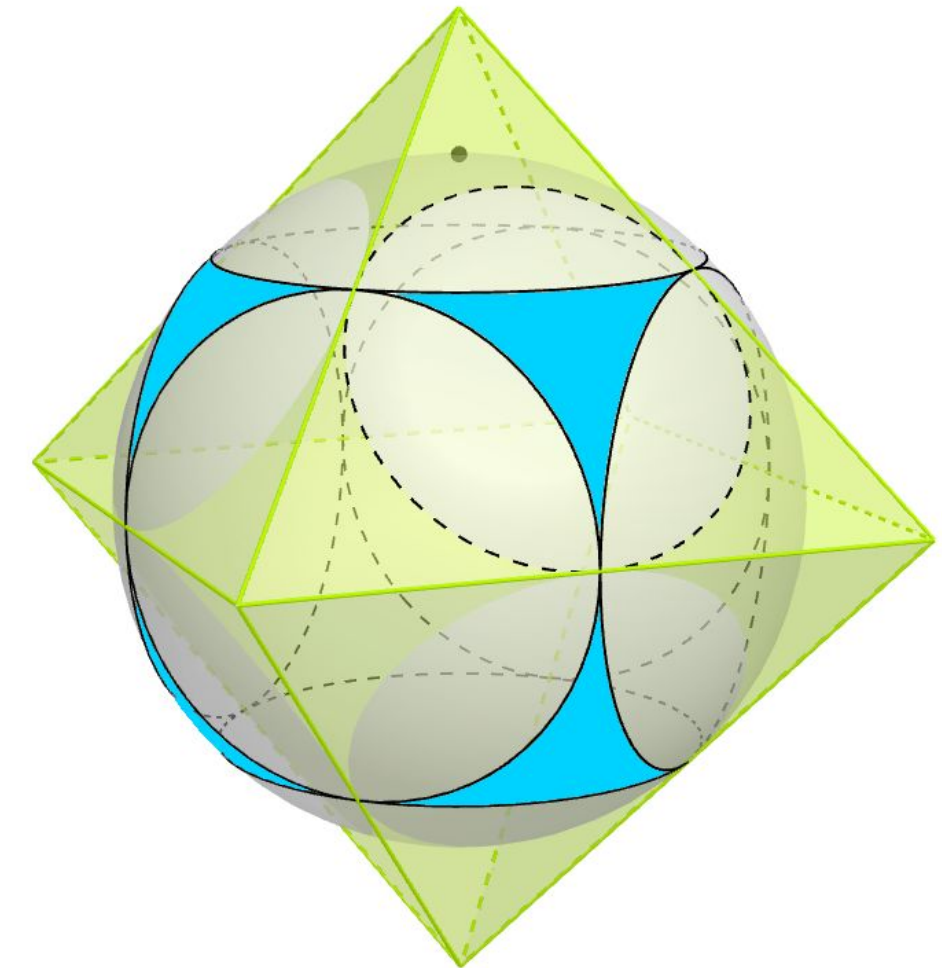
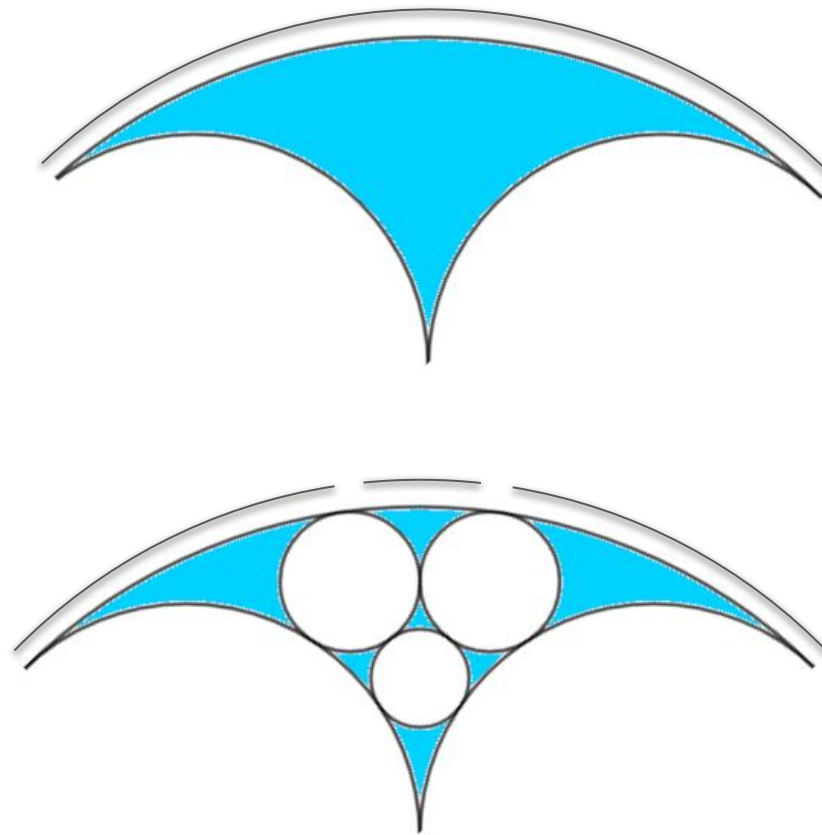
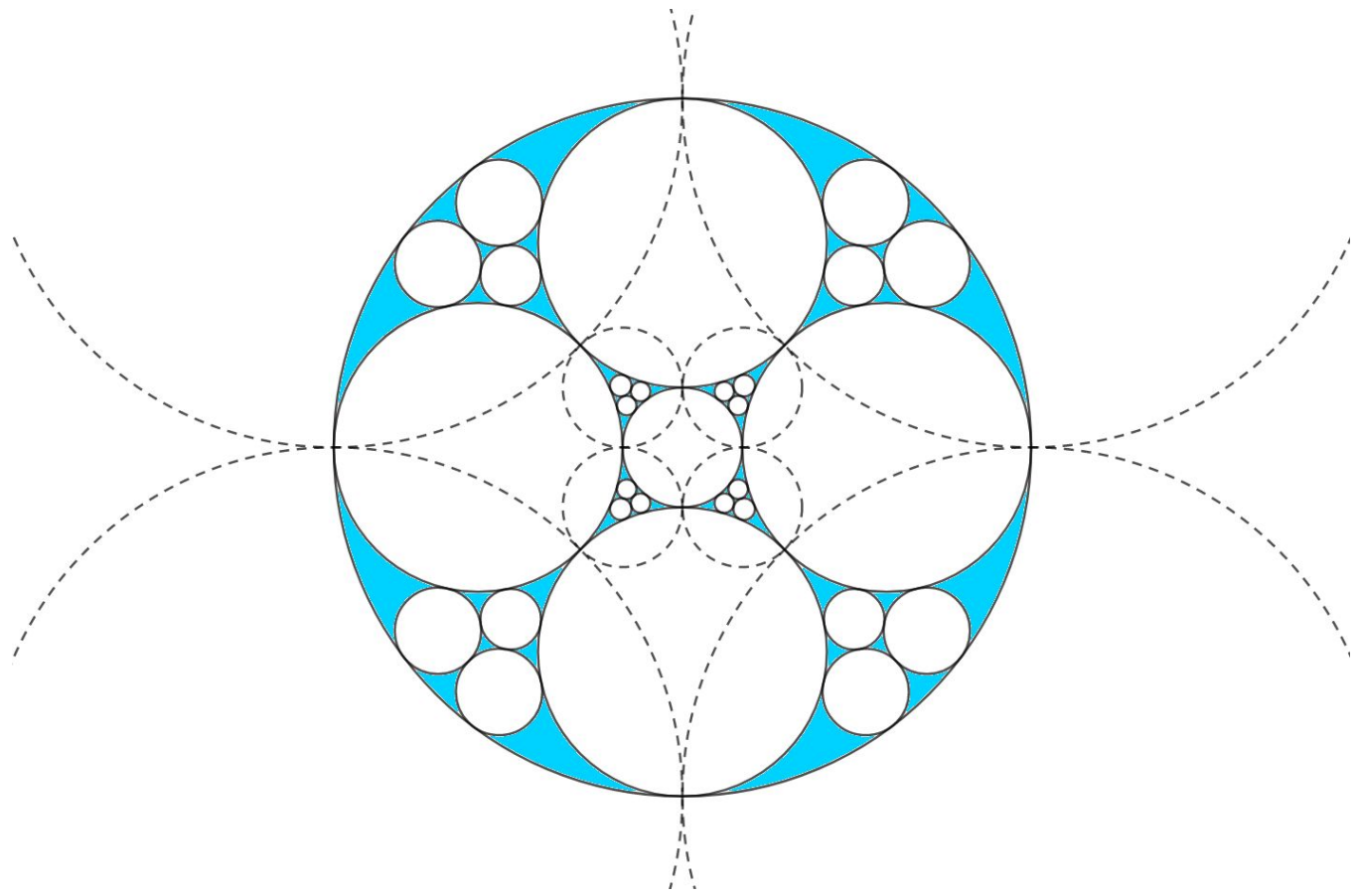
Quelles contraintes d'adjacence ?

Adjacences des subdivisions = adjacences des faces du polyèdre



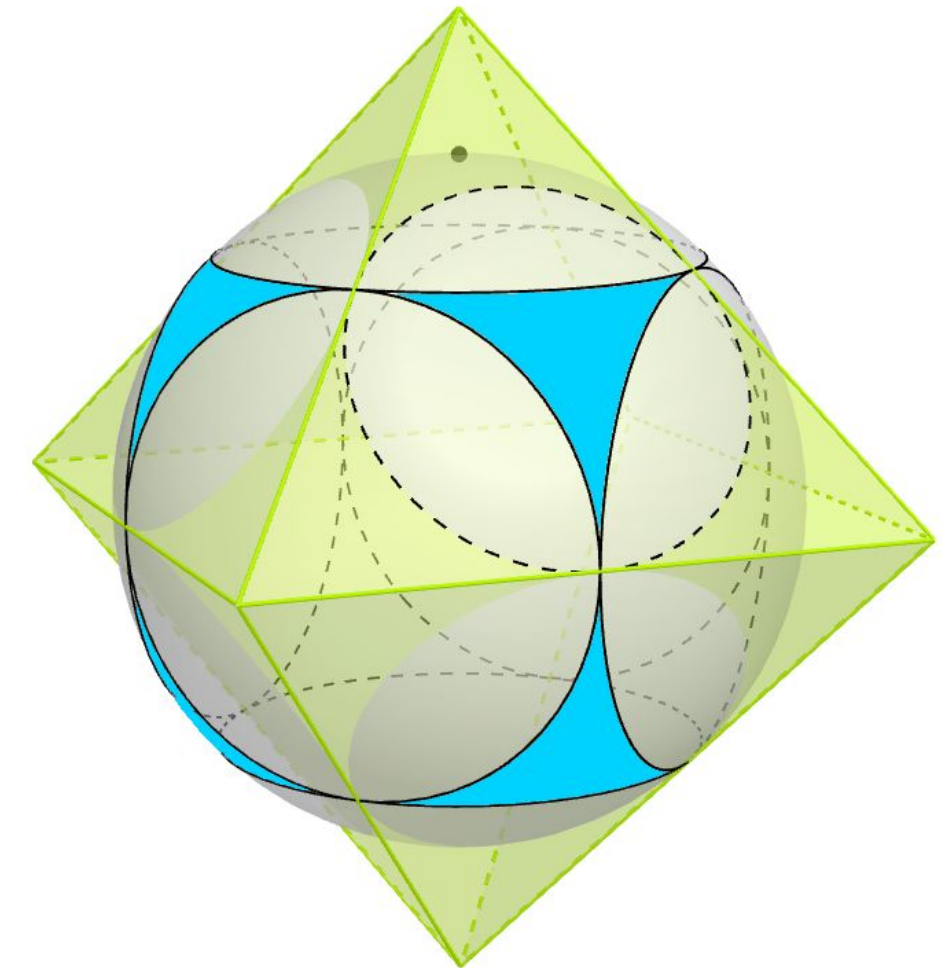
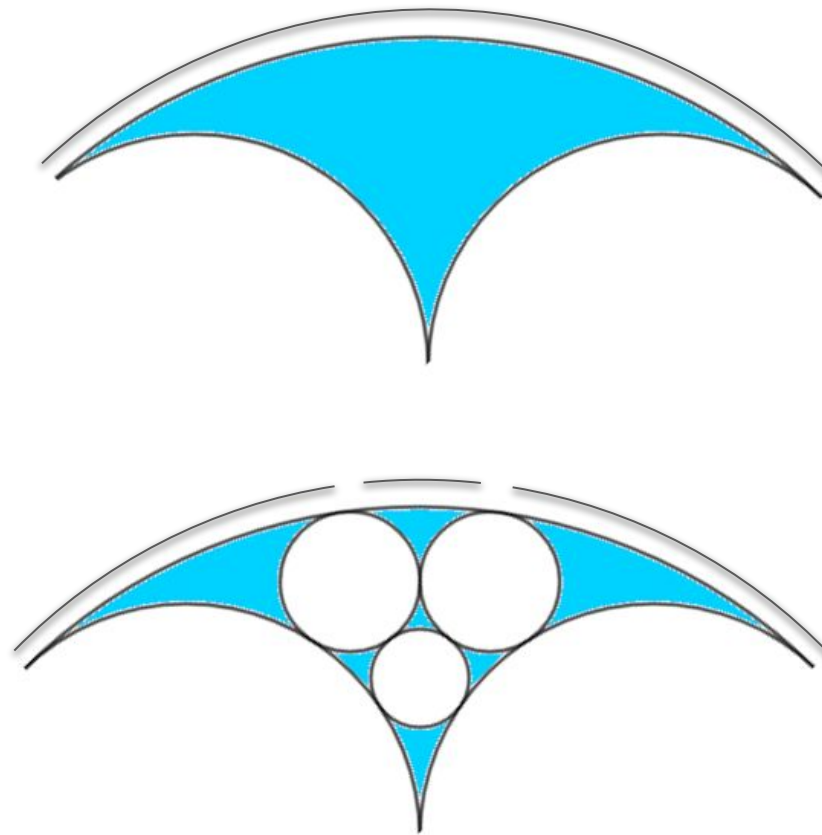
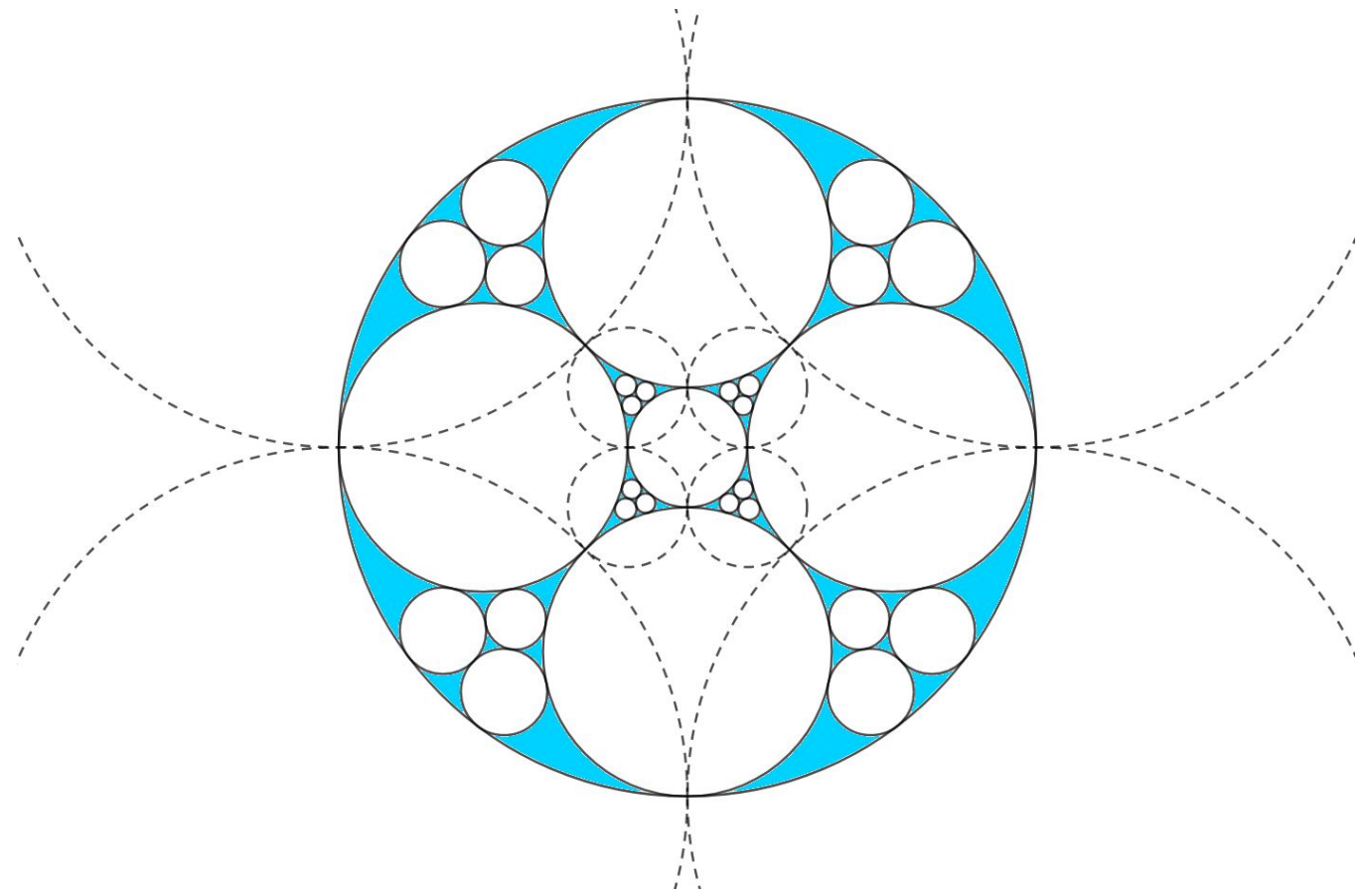
Comment se subdivisent les bords ?

Nombre de subdivisions des bords = degré du sommet associé - 1



Quelles contraintes d'incidence ?

Incidences des subdivisions = mêmes sommets représentants

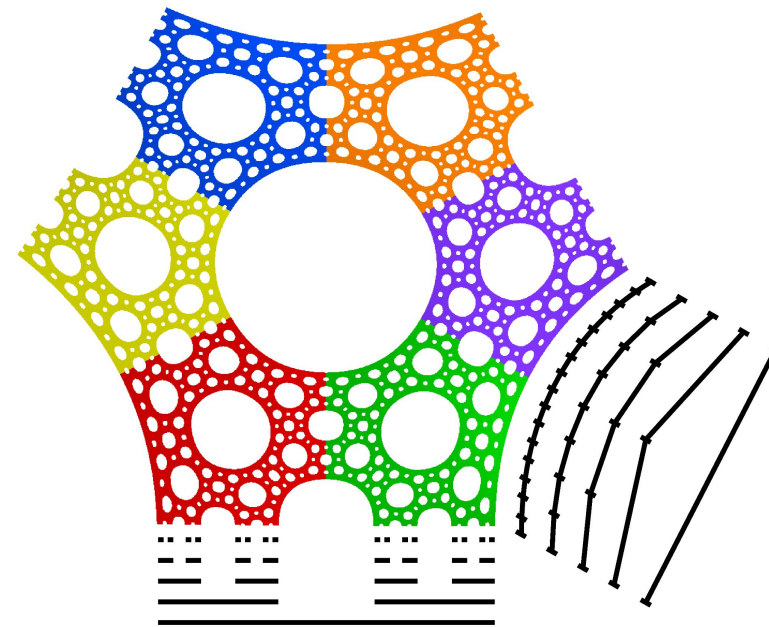


- Combien de faces ?
 - Une face fractale par face du polyèdre
- Comment sont-elles agencées ?
 - Adjacences des faces fractales = adjacences des faces du polyèdre
- Combien de bords par face ?
 - Nombre de bords = nombre de sommets
- Comment se subdivisent-elles ?
 - Nombre de subdivisions = nombre de faces du polyèdre - 1
- Quelles contraintes d'adjacence ?
 - Adjacences des subdivisions = adjacences des faces du polyèdre
- Comment se subdivisent les bords ?
 - Nombre de subdivisions des bords = degré du sommet associé - 1
- Quelles contraintes d'incidence ?
 - Incidences des subdivisions = mêmes sommets représentants

Conclusion



- Codage de la topologie des empilements de sphères en 2D
- Extraction de règles de construction
 - Construction automatique
- Pas imprimable en 2D
 - Raccords en 1 point
- Travaux futurs
 - Rendre imprimable les empilements 2D
 - Polyèdres non réguliers
 - Et en 3D ?
 - Problème inverse ?



Merci de votre attention

